# Network Virtualization

# Network Virtualization (NV)

## State of the Art

- Network Virtualization:
  State of the Art and Research Challenges
  N.M. Mosharaf Kabir Chowdhury and Raouf Boutaba,
  University of Waterloo, July 2009
  http://www.mosharaf.com/wp-content/uploads/nv-overview-commag09.pdf

# Network virtualization (NV)

- Server virtualization hides server resources.
- This practice can create problems if the data center is using traditional network architectures.
- For example, Virtual LANs (VLANs) used by VMs must be assigned to the same switch port as the physical server running the hypervisor.
- Another problem is that traffic flows differ substantially from the traditional client-server model.
- Typically, a data center has a considerable amount of traffic being exchanged between virtual servers (referred to as East-West traffic).
- These flows change in location and intensity over time, requiring a flexible approach to network resource management.
- Existing network infrastructures can respond to changing requirements related to the management of traffic flows by using Quality of Service (QoS) and security level configurations for individual flows.
- However, in large enterprises using multivendor equipment, each time a new VM is enabled, the necessary reconfiguration can be very time-consuming.

# NV – Historical perspective

- **VLAN – Virtual Local Area Network**
- **VPN – Virtual Private Network, can be classified into:**
  - Layer 1 VPN
  - Layer 2 VPN
  - Layer 3 VPN
  - Higher Layer VPNs
- **Active and Programmable networks, two separate schools of thought:**
  - Open Signaling Approach and Active Networks Approach
  - Programmability, Isolation, Provisioning
- **Overlay networks:**
  - Logical network built on top of one or more existing physical networks
  - Overlays do not require any changes to the underlying network

# NV – Business Model

With the **traditional model**: Only **one role**: ISPs

With the **NV model**: **2 roles:**

- **Infrastructure Providers (InP):**
  Deploy and actually manage the underlying physical network resources.

- **Service Providers (SP):**
  Lease resources from multiple InPs to create and deploy virtual networks by programming allocated network resources to offer end-to-end services to the end users.

For **both models:**

- **End User:**
  End users in the network virtualization model are similar to that of the existing Internet, except that the existence of multiple virtual networks from competing SPs provides them a wider range of choice.

# NV – Architecture

## NV architecture example

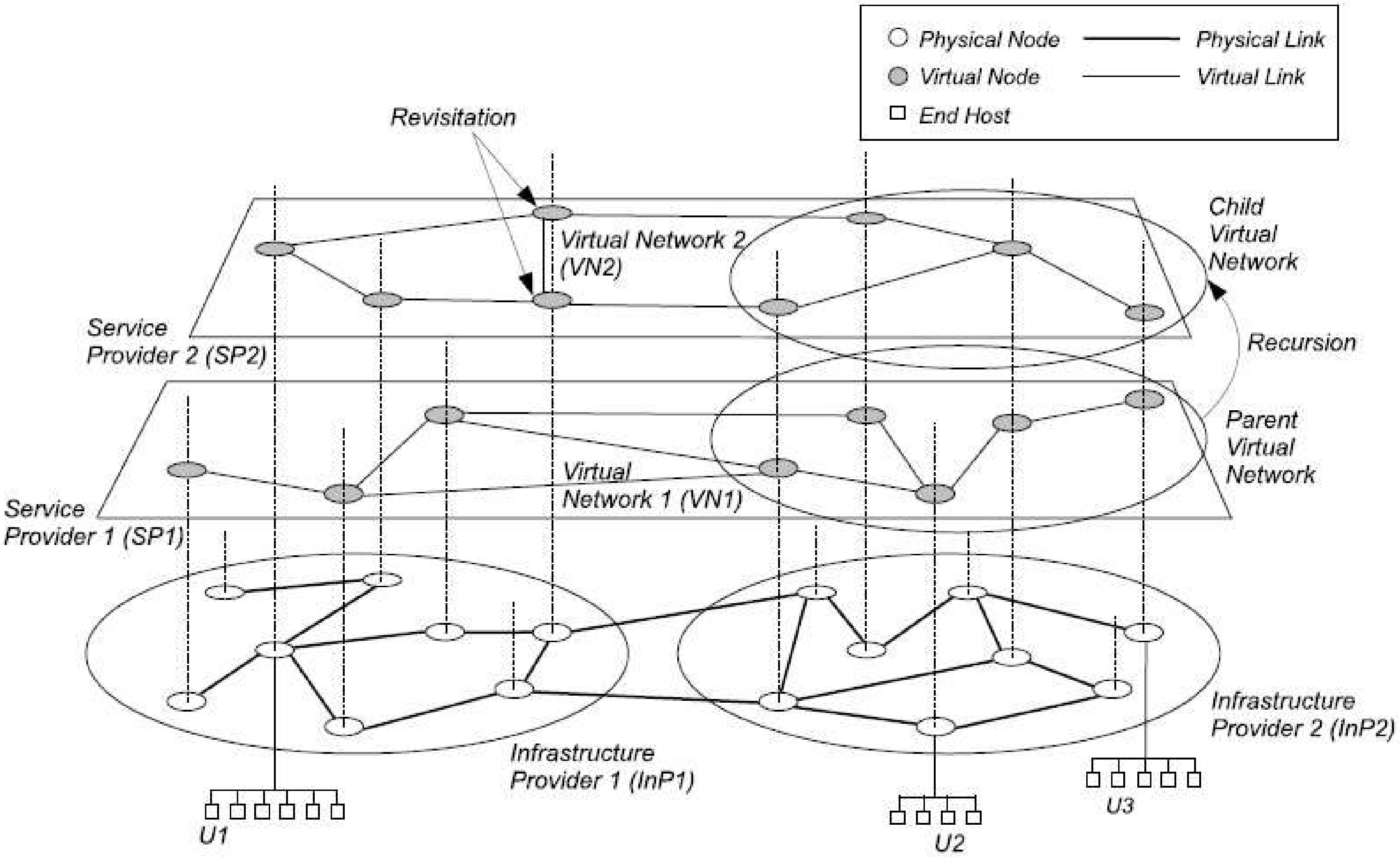



Fig. 1. Network virtualization environment

# NV – Architecture principles

- **Coexistence:** Multiple VNs from different SPs can coexist together, spanning over part or full of the underlying physical networks provided by one or more InPs. In Figure 1, VN1 and VN2 are two coexisting VNs.
- **Recursion:** When one or more VNs are spawned from another VN
- **Inheritance:** Child VNs in an NVE can inherit architectural attributes from their parents
- **Revisitation:** Revisitation allows a physical node to host multiple virtual nodes of a single VN.

# NV – Design goals – 1

- **Flexibility:** Each SP should be free to implement arbitrary network topology, routing and forwarding functionalities, and customized control protocols independent of the underlying physical network and other coexisting VNs.
- **Manageability:** By separating SPs from InPs, NV will modularize network management tasks and introduce accountability at every layer of networking
- **Scalability:** Coexistence of multiple networks
- **Isolation:** NV must ensure isolation between coexisting VNs to improve fault-tolerance, security, and privacy.
- **Stability and Convergence:** Errors and misconfigurations in the underlying physical network can also destabilize an NVE. Moreover, instability in the InPs (e.g., routing oscillation) can lead to instability of all the hosted VNs.

# NV – Design goals – 2

- **Programmability:** To ensure flexibility and manageability, programmability of the network elements is an indispensable requirement. Only through programmability, SPs can implement customized protocols and deploy diverse services.

- **Legacy Support:** Legacy support or backward compatibility has always been a matter of deep concern while deploying any new technology.

- **Heterogeneity:** Heterogeneity in the context of network virtualization comes mainly from two fronts: first, heterogeneity of the underlying networking technologies (e.g., optical, wireless, and sensor); second, each end-to-end VN, created on top of that heterogeneous combination of underlying networks, can also be heterogeneous. SPs must be allowed to compose and run crossdomain end-to-end VNs without the need for any technology specific solutions.

# NV – Projects

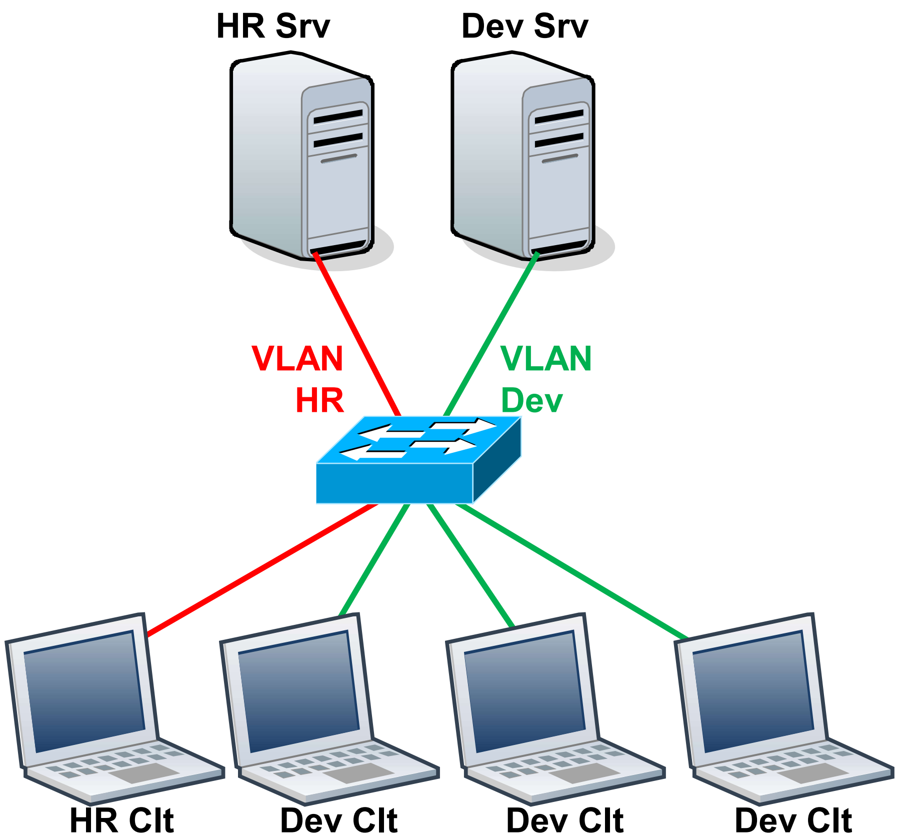
**Characteristics of different network virtualization projects over the years**

| Project | Architectural Domain | Networking Technology | Layer of Virtualization | Level of Virtualization |
|---|---|---|---|---|
| **VNRMS** | Virtual network management | ATM/IP | | Node/Link |
| **Tempest** | Enabling alternate control architectures | ATM | Link | |
| **NetScript** | Dynamic composition of services | IP | Network | Node |
| **Genesis** | Spawning virtual network architectures | | Network | Node/Link |
| **VNET** | Virtual machine Grid computing | | Link | Node |
| **VIOLIN** | Deploying on-demand value-added services on IP overlays | IP | Application | Node |
| **X-Bone** | Automating deployment of IP overlays | IP | Network | Node/Link |
| **PlanetLab** | Deployment and management of overlay-based testbeds | IP | Application | Node |
| **UCLP** | Dynamic provisioning and reconfiguration of lightpaths | SONET | Physical | Link |
| **AGAVE** | End-to-end QoS-aware service provisioning | IP | Network | |
| **GENI** | Creating customized virtual network testbeds | Heterogeneous | | |
| **VINI** | Evaluating protocols and services in a realistic environment | | Link | |
| **CABO** | Deploying value-added end-to-end services on Heterogeneous shared infrastructure | Heterogeneous | | Full |

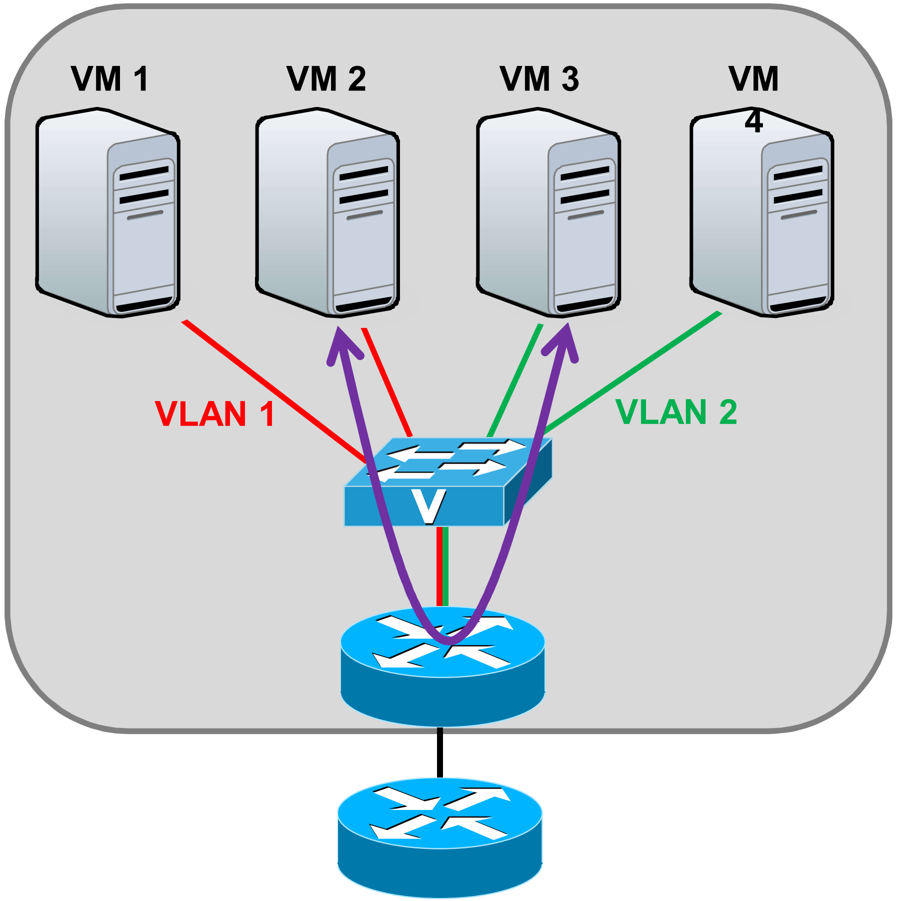
# NV – External / Internal

## How it improves network resource usage?

### External NV example




### Internal NV example



# NV – Research challenges

- Interfacing
- Signaling and Bootstrapping
- Resource and Topology Discovery
- Resource Allocation, provisioning
- Admission Control and Usage Policing
- Virtual Nodes and Virtual Links
- Naming and Addressing
- Mobility Management
- Monitoring, Configuration, and Failure Handling
- Security and Privacy
- Interoperability Issues
- Network Virtualization Economics

# NV – Keep in mind!

- Network virtualization stands at a unique point in the virtualization design space.
  - **On the one hand**, it is necessary to have a virtualized network to interconnect all other virtualized appliances to give each of the virtual entities a complete semblance of their native counterparts.
  - **On the other hand**, after enjoying years of rapid growth, the progress of the Internet and networking in general has come to a standstill. Most researchers now agree that a redesign is a bare necessity, not luxury. Network virtualization can take a leading role in this scenario to promote innovation through disruptive technologies.

# Network Function Virtualization (NFV) – Values of NFV

Some of the values to the NFV concept:

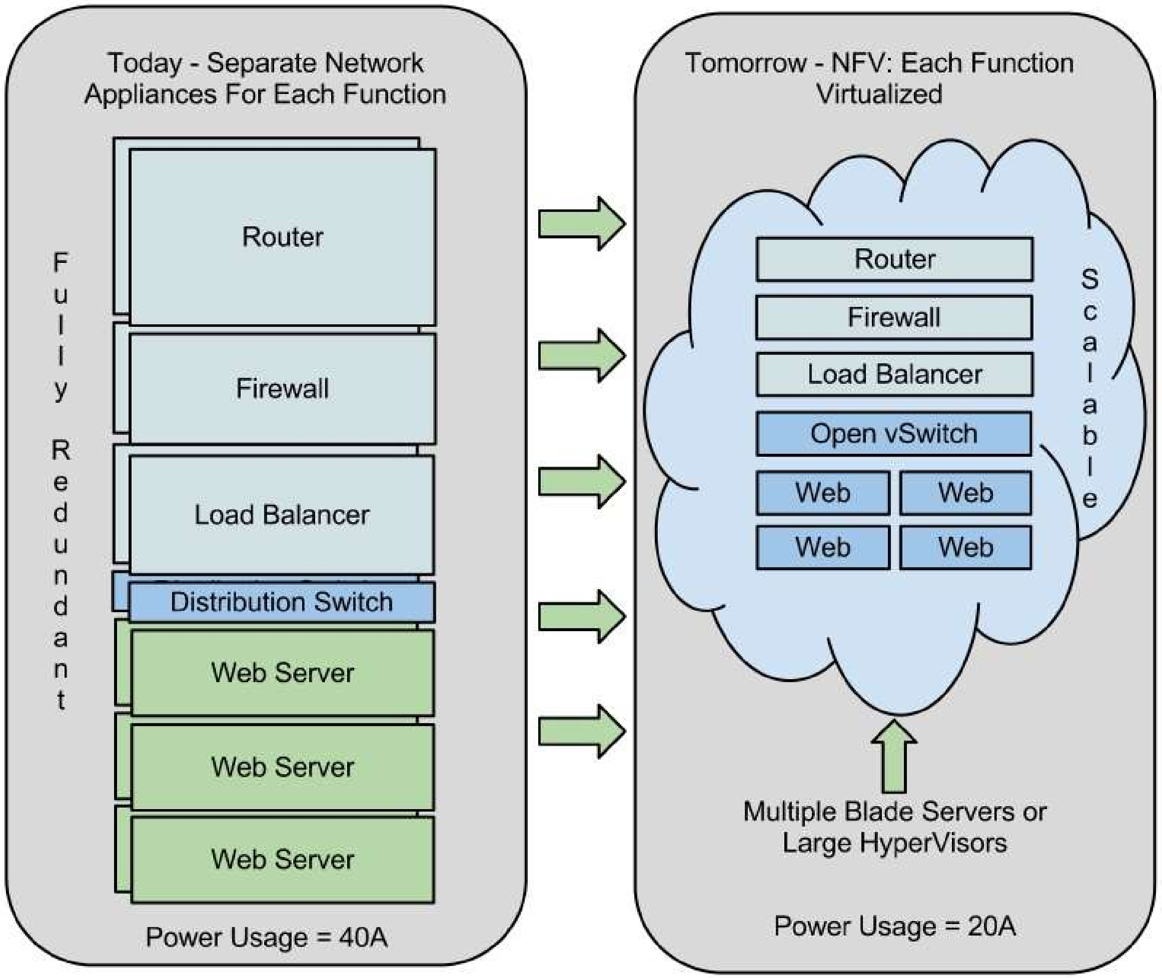
- Speed, agility, and cost reduction

By centralizing designs around commodity server hardware, network operators can:

- Do a single PoP/Site design based on commodity compute hardware
- Utilize resources more effectively
- Deploy network functions without having to send engineers to each site
- Achieve Reductions in OpEX and CapEX
- Achieve Reduction of system complexity

# NFV

## Actual architecture vs. NFV architecture



# Virtual network components – 1


Virtualized network components already exist:

- Cisco:
  - Nexus 1000-V
  - Virtual Security Gateway (VSG) for the Nexus 1000-V
- VMware:
  - vShield Edge (Firewall and VPN)
  - vShield Endpoint (security compliance and data protection solutions)
- Extreme Networks: XNV network hypervisor
- Vyatta:
  - Enterprise-grade virtual network router, firewall and VPN solution to install directly on physical servers and turn them into routers.
- …

# Virtual network components – 2

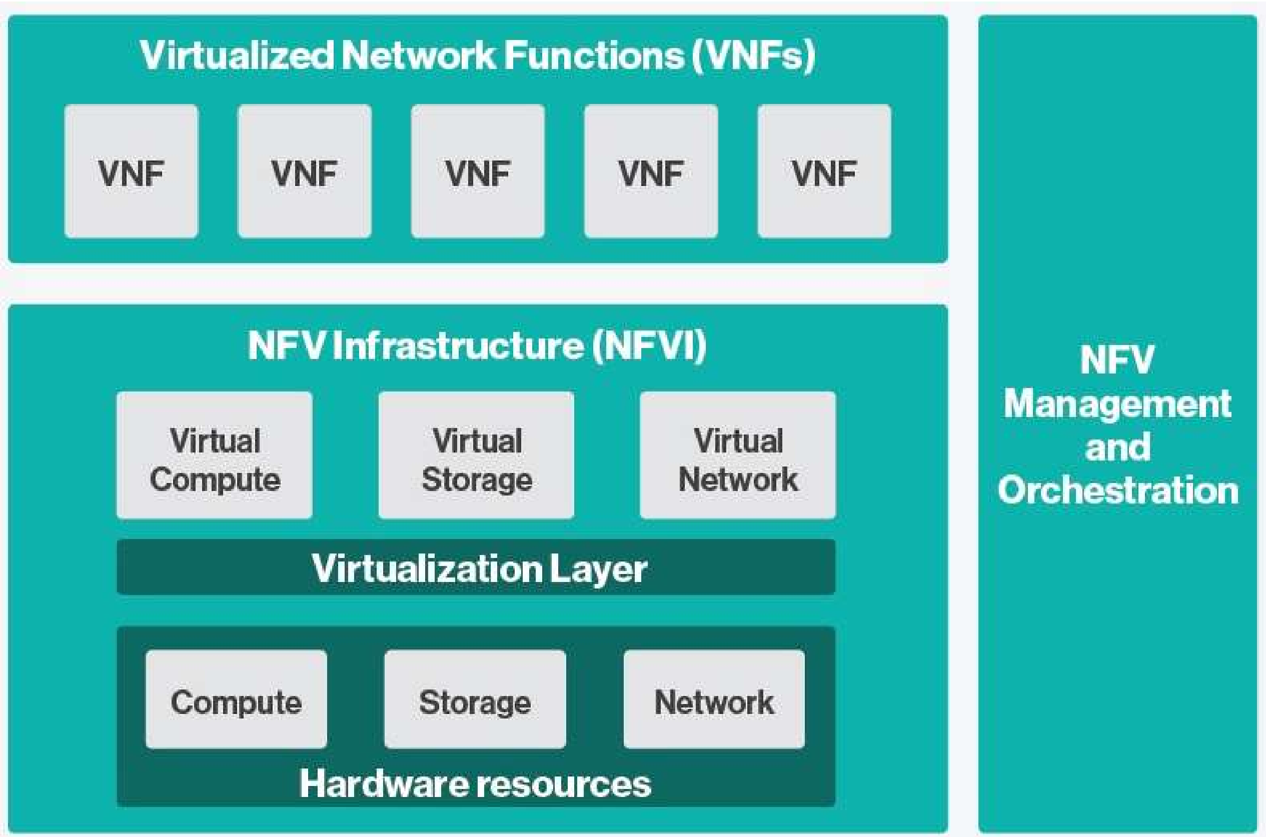
## Open vSwitch

- Production quality OpenSource software switch for virtualized server environments
- It forwards traffic between different VMs on the same physical host
- It forwards traffic between VMs and the physical network
- Supported management interfaces: sFlow, NetFlow, RSPAN, CLI
- Programmable and controlled using OpenFlow and OVSDB

➢ More information on: https://openvswitch.org/

# NFV vs VNF

- NFV is an overarching concept, while a VNF is building block within ETSI's current NFV framework.



# Network virtualization – Glossary

## Glossary

- https://whatis.techtarget.com/glossary/Virtualization
- https://whatis.techtarget.com/glossary/Network-Administration

## References

- https://www.mosharaf.com/wp-content/uploads/nv-overview-commag09.pdf
- https://searchservervirtualization.techtarget.com/tip/QA-Network-virtualization-benefits-and-challenges
- https://searchservervirtualization.techtarget.com/tip/Network-virtualization-technology-FAQ
- https://www.techtarget.com
- https://www.vmware.com/topics/network-functions-virtualization-nfv
- http://searchsdn.techtarget.com/definition/network-functions-virtualization-NFV
- https://www.reddit.com/r/homelab/comments/1bqtx87/do_i_really_need_a_hypervisor/

# Cloud networking

# Cloud networking – 1

- Ultimately, the goal of cloud computing is to create a fluid pool of resources across servers and data centers that enable users to access stored data and applications on an as-needed basis.

- Cloud computing networks, therefore, have two missions:
  - to enable the movement of that pool as a single virtual resource
  - to connect users to these resources regardless of location.

# Cloud networking – 2

**To make that happen, cloud computing networks must be able to:**

- Burst up and turn down bandwidth on demand
- Provide extremely low latency throughput among storage networks, the data center and the LAN
- Allow for non-blocked connections between servers to enable automated movement of virtual machines (VMs)
- Function within a management plane that stretches across enterprise and service provider networks
- Provide visibility despite this constantly changing environment

# Cloud networking – 3

- Cloud computing networks can be seen as three interdependent structures:
  - the front-end, which connects users to applications;
  - a horizontal aspect, which interconnects physical servers and the movement of their VMs;
  - and storage networks.
- The larger cloud network can be built as either a layer 2 or a layer 3 network.

# Cloud networking – 4

- Read more: Cloud networking: Inter-networking data centers and connecting users, published in Feb 2010
- http://searchnetworking.techtarget.com/tip/Cloud-networking-Inter-networking-data-centers-and-connecting-users
  - ...
  - Cloud networking within the data center: Addressing loss and latency
  - Cloud computing network: Flat networks mean fewer interfaces along the way
  - If you can't go flat: Managing trunk and port connections in layered cloud networks
  - Inter-networking data centers for cloud networking
  - Cloud networking traffic management: Start with user connections
  - Address management for virtual components in a cloud network
  - Connecting public and private clouds

# Cloud networking – 5

- Read more: Evolutionary Designs for Cloud Networking, published in Jan 2009
- https://www.arista.com/en/
- https://blogs.arista.com/blog?p=49
- ...
  - Access Layer is changing
  - ...
  - Cloud performance uniformity is key
  - ...
  - Introducing cloud leaf and spine
  - ...
  - Virtualization considerations in a cloud network
  - ...

# Cloud networking – 6

- Read more: Cloud computing – A primer – Part 1: Models and technologies, published in Sep 2009
  - ...
  - How is cloud computing different from hosted services?
  - Virtualization and its effect on cloud computing
  - VM migration: An advantage of virtualization
  - Major models in cloud computing
  - Platform as a Service
- Infrastructure as a Service
- Public, private and internal clouds
- When does cloud computing make sense?
- Cloud computing infrastructure
- Storage infrastructure
- Cloud computing: Effect on the network
- New protocols for Data-Center networking
- Virtualized network equipment functions
- Management
- Cloud computing: Common myths
- Cloud computing: Gaps and concerns
- Conclusion

# Cloud networking – 7

- Read more: Cloud computing – A primer – Part 2: Infrastructure and implementation topics, published in Sep 2009
  - ...
  - Network infrastructure
  - Data-Center infrastructure extension – IaaS
  - PaaS and SaaS infrastructure
  - Virtualization and its demands on switching
  - IaaS private clouds
  - Layer 2 versus Layer 3 connectivity for cloud networks
  - Cloud federation
  - Security
  - Virtualization and security
  - Standards bodies involved in cloud computing
  - Some perspectives on cloud computing
  - Conclusion

# Cloud networking – Solutions

## Overlay network:

- Use of server virtualization and cloud computing is stressing the network infrastructure in several ways:
  - Server Virtualization increases demands on switch MAC address tables
  - Multi-tenancy and vApps driving the need for more than 4K VLANs
  - Static VLAN trunk provisioning doesn't work well for Cloud Computing and VM mobility
  - Limited reach of VLANs using STP constrains use of compute resources

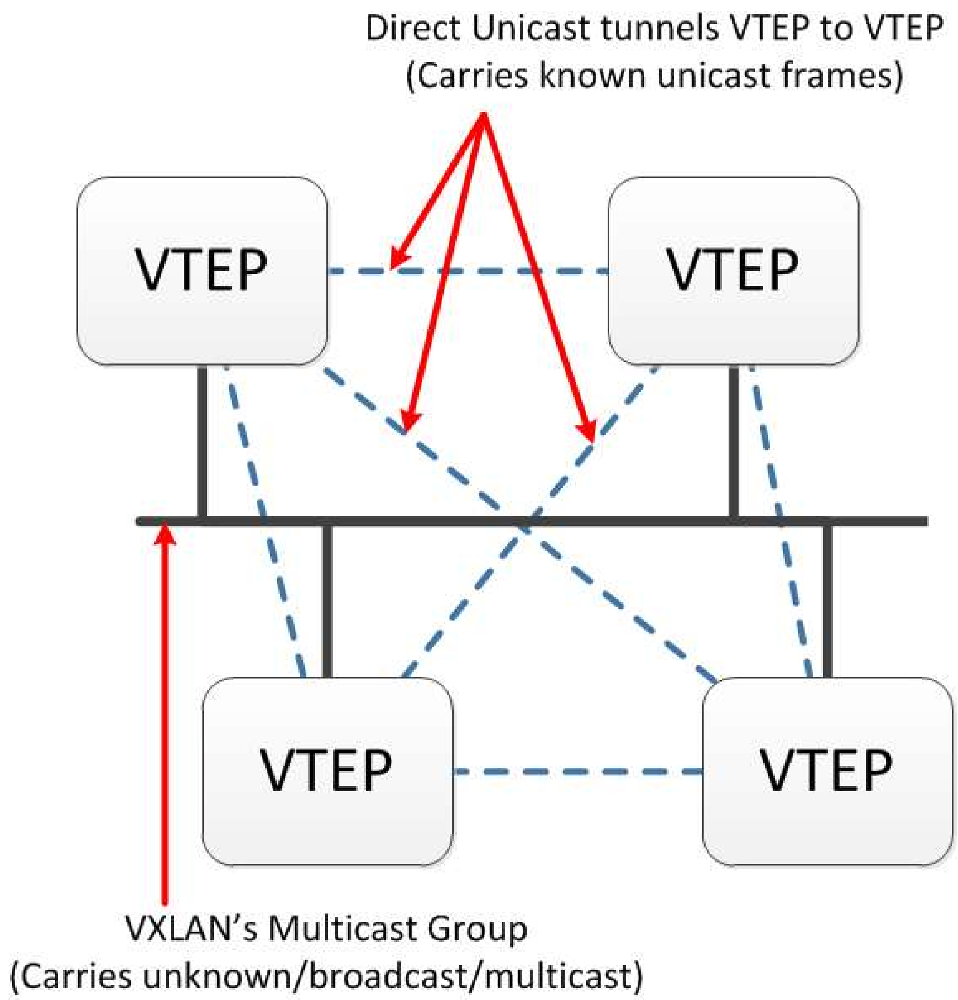
- VXLAN: Virtual eXtensible Local Area Network
- NVGRE: Network Virtualization using Generic Routing Encapsulation
- STT: Stateless Transport Tunneling

# Cloud networking – VXLAN

- Proposed by Cisco and VMware
- Increases the number of available IDs to 16 million with a 24-bit VXLAN Network Identifier (VNI)
  - Each VNI may contain up to 4094 VLANs
- Encapsulation of a virtual Layer 2 network (only visible to VMs) over a physical Layer 3 network with UDP packets → MAC Over IP/UDP
- No changes on the VM software. A VTEP (Virtual Tunnel End Point), which resides within the server hypervisor, assigns VNIs to VMs.

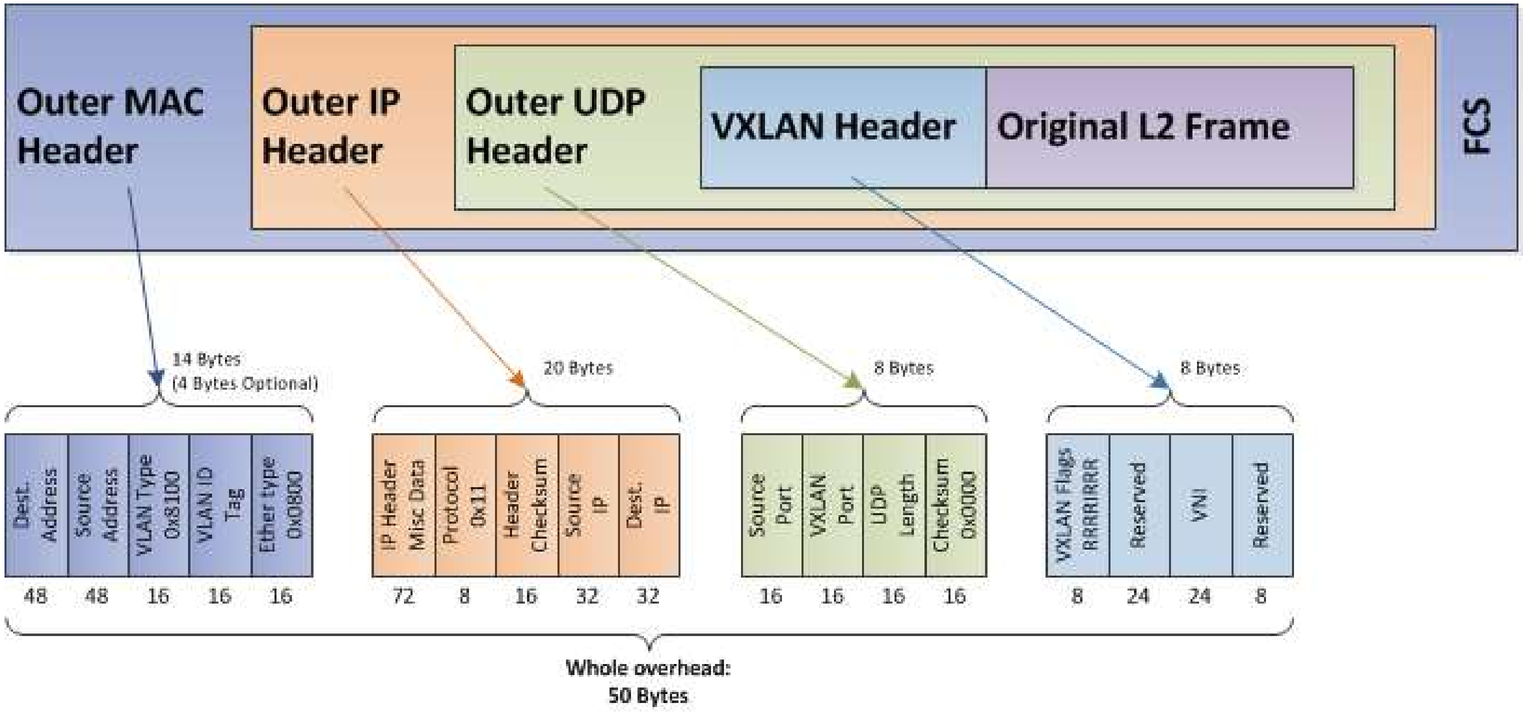
# Cloud networking – VXLAN

## VXLAN flow categorization



# Cloud networking – VXLAN

## VXLAN encapsulation

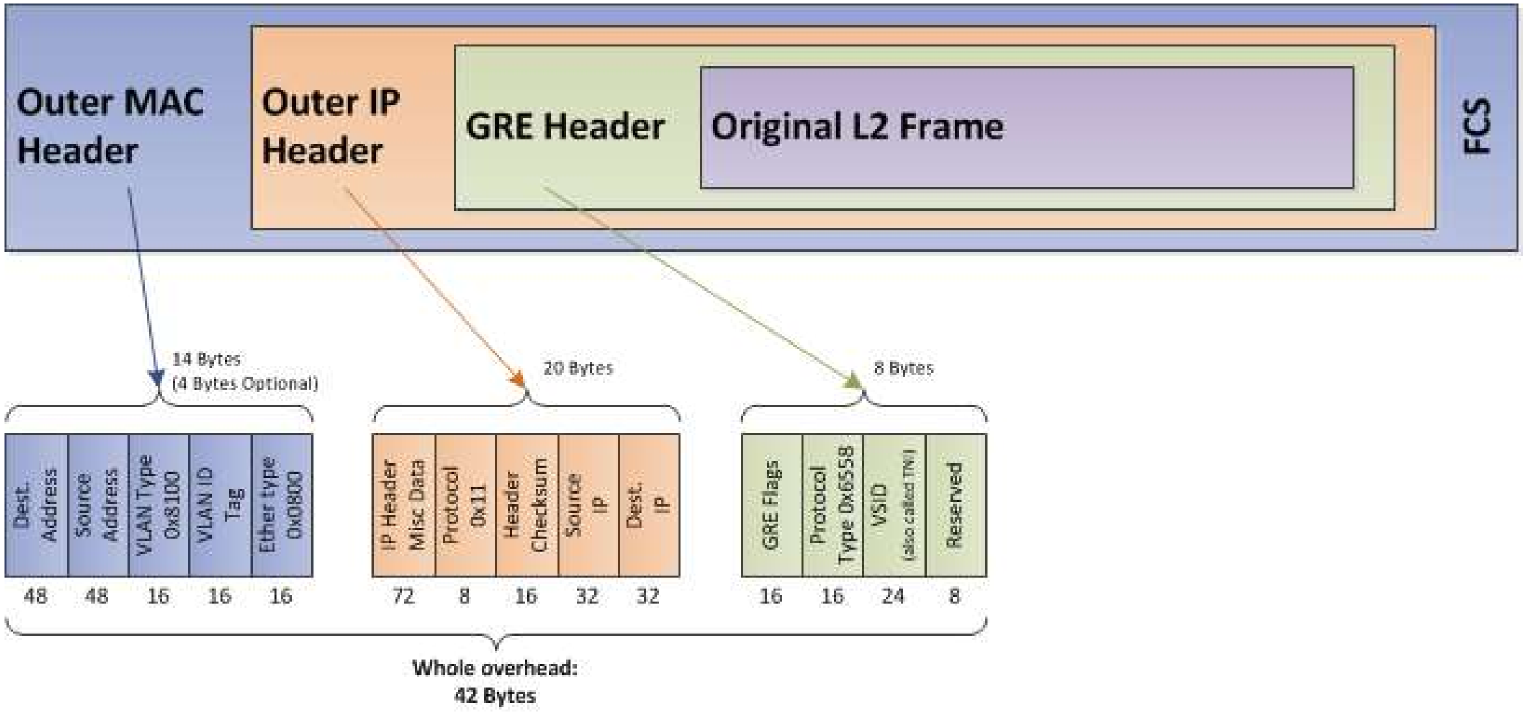


# Cloud networking – NVGRE

- Proposed by Microsoft, Intel, HP and Dell …
- Use encapsulation and tunneling to create large numbers of VLANs for subnets that can extend across dispersed data centers
  - The NVGRE proposal uses GRE to create an isolated virtual L2 network
- Specifies a 24-bit Tenant Network Identifier (TNI), and allows up to 16 million networks (Also called Virtual Subnet ID – VSID)
- Each TNI is associated with an individual GRE tunnel and sends packets with IP multicast
- The NVGRE end points are typically implemented in the server hypervisor

# Cloud networking – NVGRE

## NVGRE encapsulation



# Cloud networking – STT

- Proposed by Nicira, Broadcom, Rackspace, eBay, Intel and Yahoo
- Use a TCP-like header to benefit from TSO (TCP Segmentation Offload)
  - The segmentation is performed by the NIC, which saves CPU resources and improve performance
- Use of a “Context-ID” of 64 bits to separate networks
- May be problematic with Firewall:
  - But is that the STT frames should they really go through a firewall?

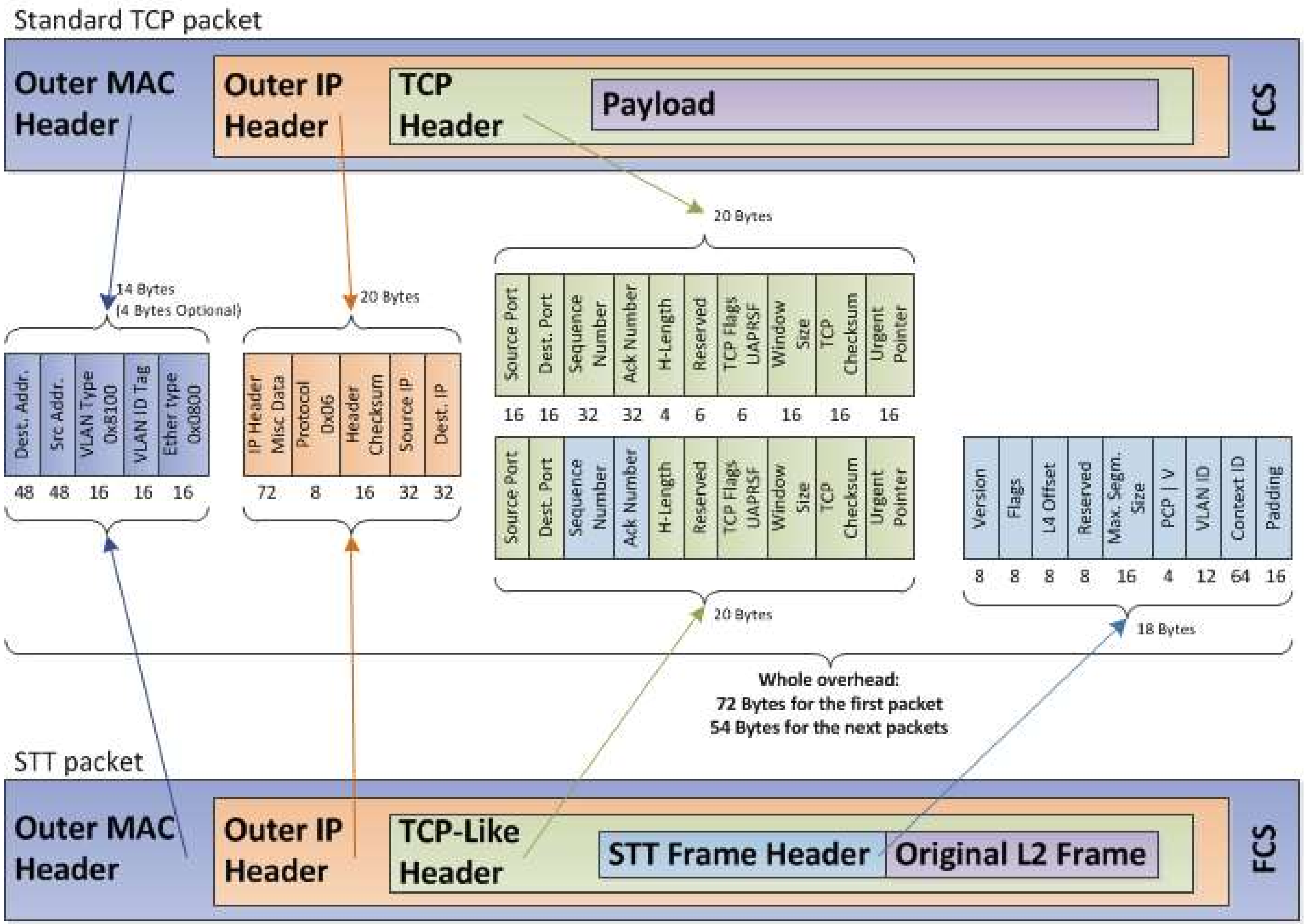
```
                   +------------+   +-----------+   +-----------+
                   | IP Header  |   |IP Header  |   |IP header  |
+------------+     +------------+   +-----------+   +-----------+
|STT Frame   |     |TCP-like    |   |TCP-like   |   |TCP-like   |
| Header     |     | header     |   | header    |   | header    |
+------------+     +------------+   +-----------+   +-----------+
|            | --->| STT Frame  |   |Next part  |...|Last part  |
|Payload     |     |  Header    |   |of Payload |   |of Payload |
.            .     +------------+   |           |   |           |
.            .     |            |   |           |   |           |
.            .     |  Start of  |   |           |   |           |
+------------+     |  Payload   |   |           |   +-----------+
                   +------------+   +-----------+

Original data          STT Frame is segmented and transmitted as
frame is encapped          a set of TCP segments (MAC
with STT Header                headers not shown)

             Figure 2: STT Frame Fragments and Encapsulation
```

# Cloud networking – STT

## Standard TCP packet vs. STT packet



# Cloud networking – Glossary

## References


- http://searchnetworking.techtarget.co.uk/tutorial/Cloud-computing-network-primer
- http://searchnetworking.techtarget.com/tip/Cloud-networking-Inter-networking-data-centers-and-connecting-users
- http://www.aristanetworks.com/en/blogs/?p=49
- http://searchnetworking.techtarget.com/tip/VXLAN-standard-primer-Extended-VLANs-long-distance-VM-migration
- http://searchnetworking.techtarget.com/tip/VXLAN-standard-primer-Extended-VLANs-long-distance-VM-migration
- http://tools.ietf.org/html/draft-davie-stt-01
- http://www.techtarget.com

# VM Migration

# VM Migration

Many words to talk about the same thing:

- VM Migration, Live Migration, VM Mobility, vMotion (VMware feature), …

What is live migration?

- Migration of a virtual machine is simply **moving the VM** running **on** a physical machine (let's call it **source node**) **to** another physical machine (let's call it **target node**).
- The trick is to do this, while the VM is running on the source node, and without disrupting any active network connections even after the VM is moved to the target node.
- It is considered "live", since the original VM is running, while the migration is in progress. Huge benefit of doing the live migration is the very small downtime in the order of milli-seconds.

# VM Migration – How to achieve?

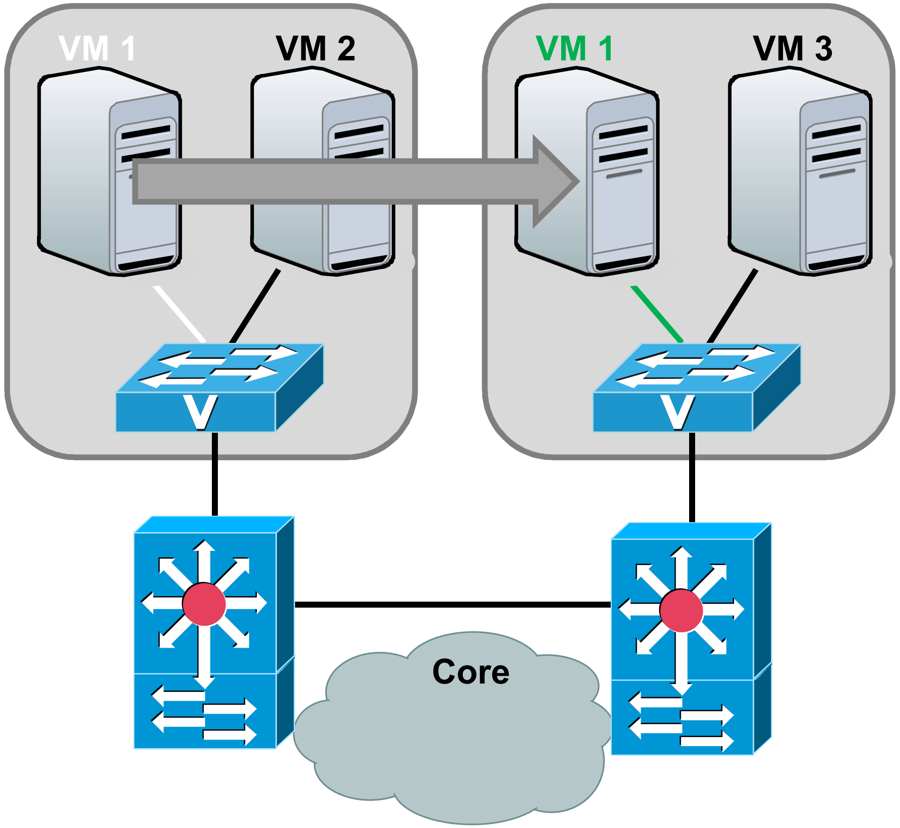
- Migrating CPU state
- Migrating memory content
  - Doing iterative copying of the memory contents
  - When the “delta” memory is small:
    - Source node is paused
    - Delta memory is copied
    - VM is resumed on target node
- Migrating storage content
  - Centralized storage is used to reduce required time
- Migrating network
  - Simple if all the nodes are in the same IP subnet
    - After migration, an ARP broadcast is sent
    - Transparent for Transp. layer. TCP connections survive to migration
    - This doesn’t work if the VMs have to cross subnets»

# VM Migration – Benefits and FAQ

- Benefits of live migration
  - Resource management in cloud computing
    - To save energy
    - To save cost
    - For load balancing
    - For maintenance purpose
    - …

- Frequently asked questions about live migration
  - What is the usual VM downtime?
  - What is the total migration time?
  - How is live migration different from suspend/resume?
  - How is live migration different from migrating to the cloud?

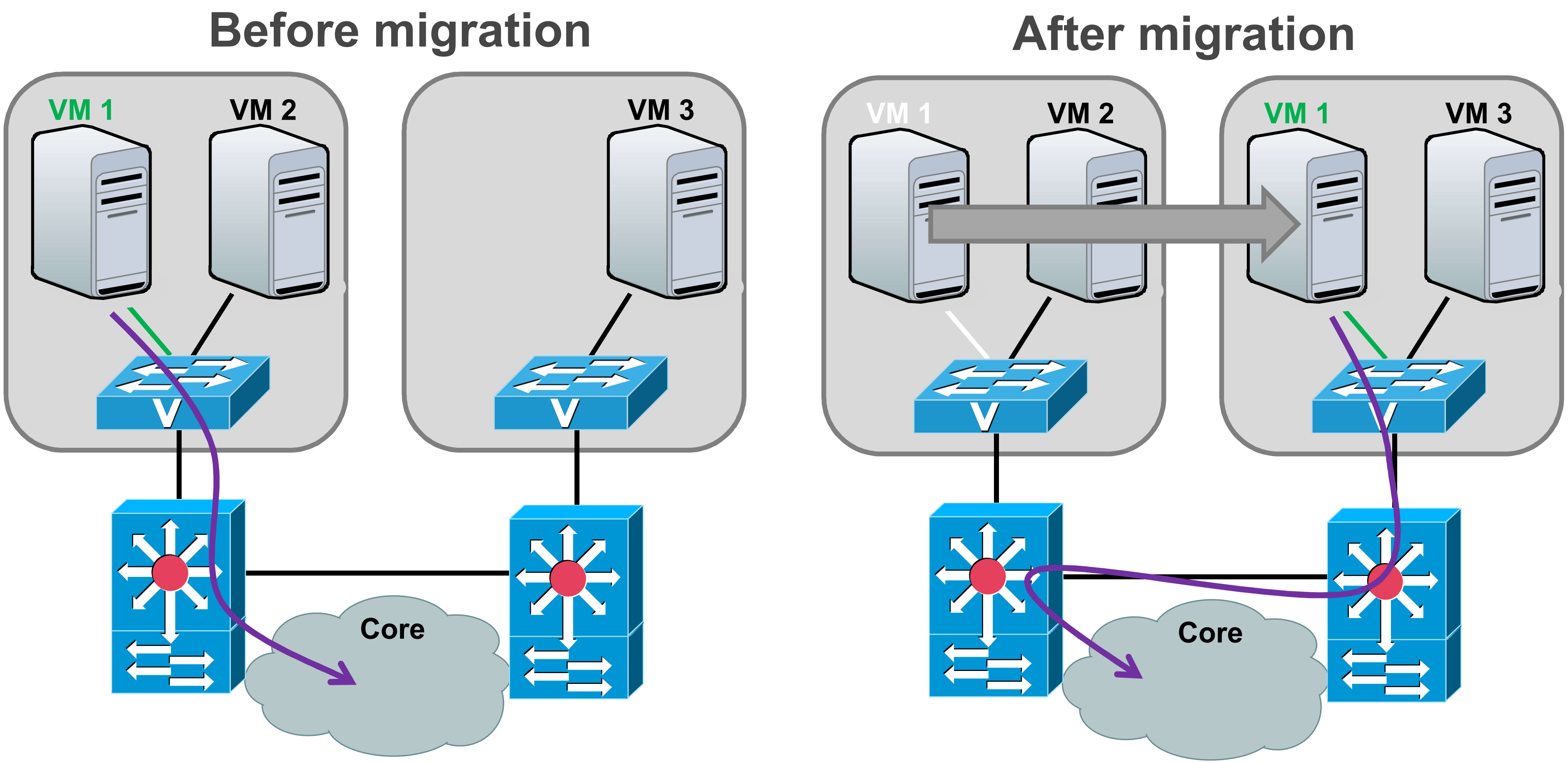
# VM Migration – Flow implications – 1



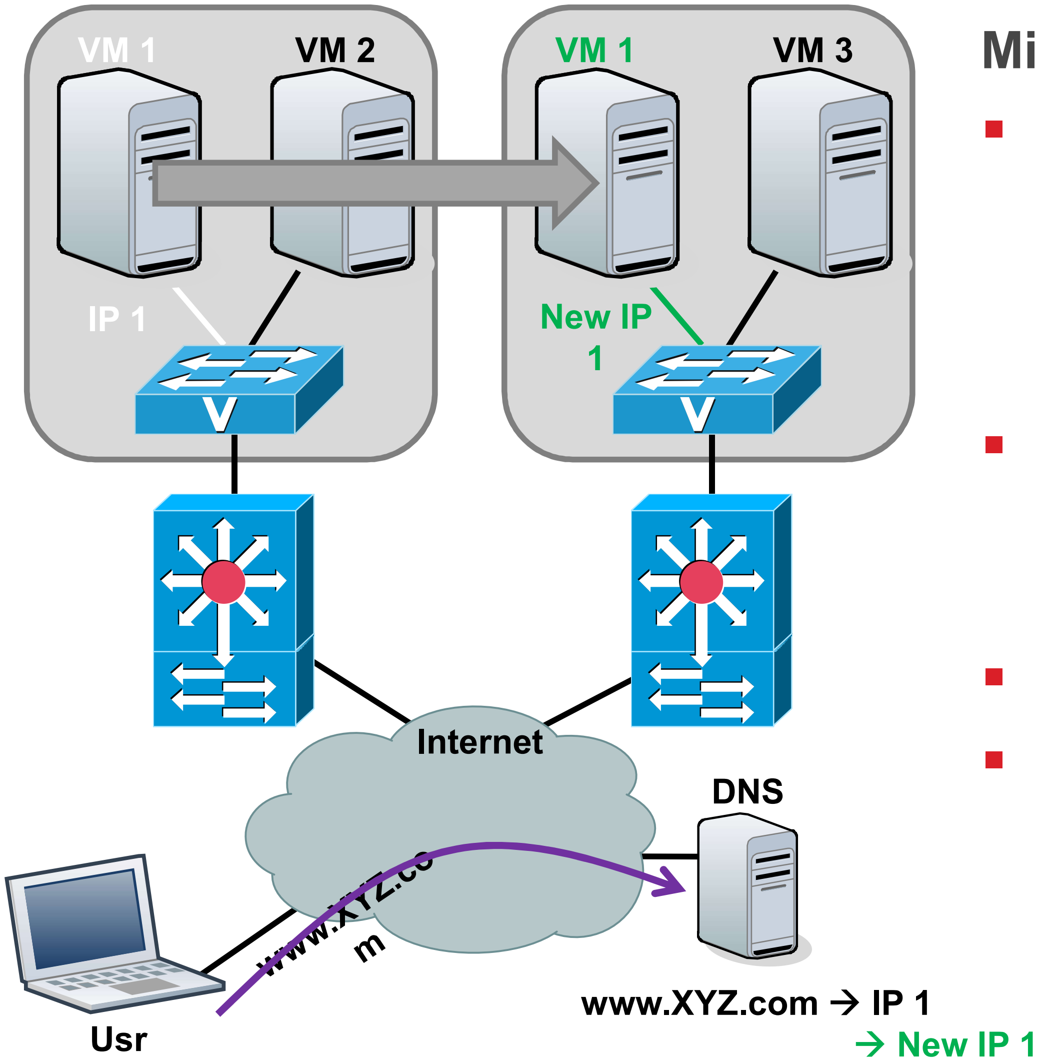

**Migration on same L2 broadcast domain:**

- ARP broadcast sent to update switch ARP-tables
- Implications when using the same-VLAN:
  - A trunking link connecting the physical server with a switch is needed.
  - All servers have to have the same set of VLANs.
  - All intermediate switches have to participate in all those VLANs.
- The server and switch configurations have to be synchronized

# VM Migration – Flow implications – 2



# VM Migration – Flow implications – 3

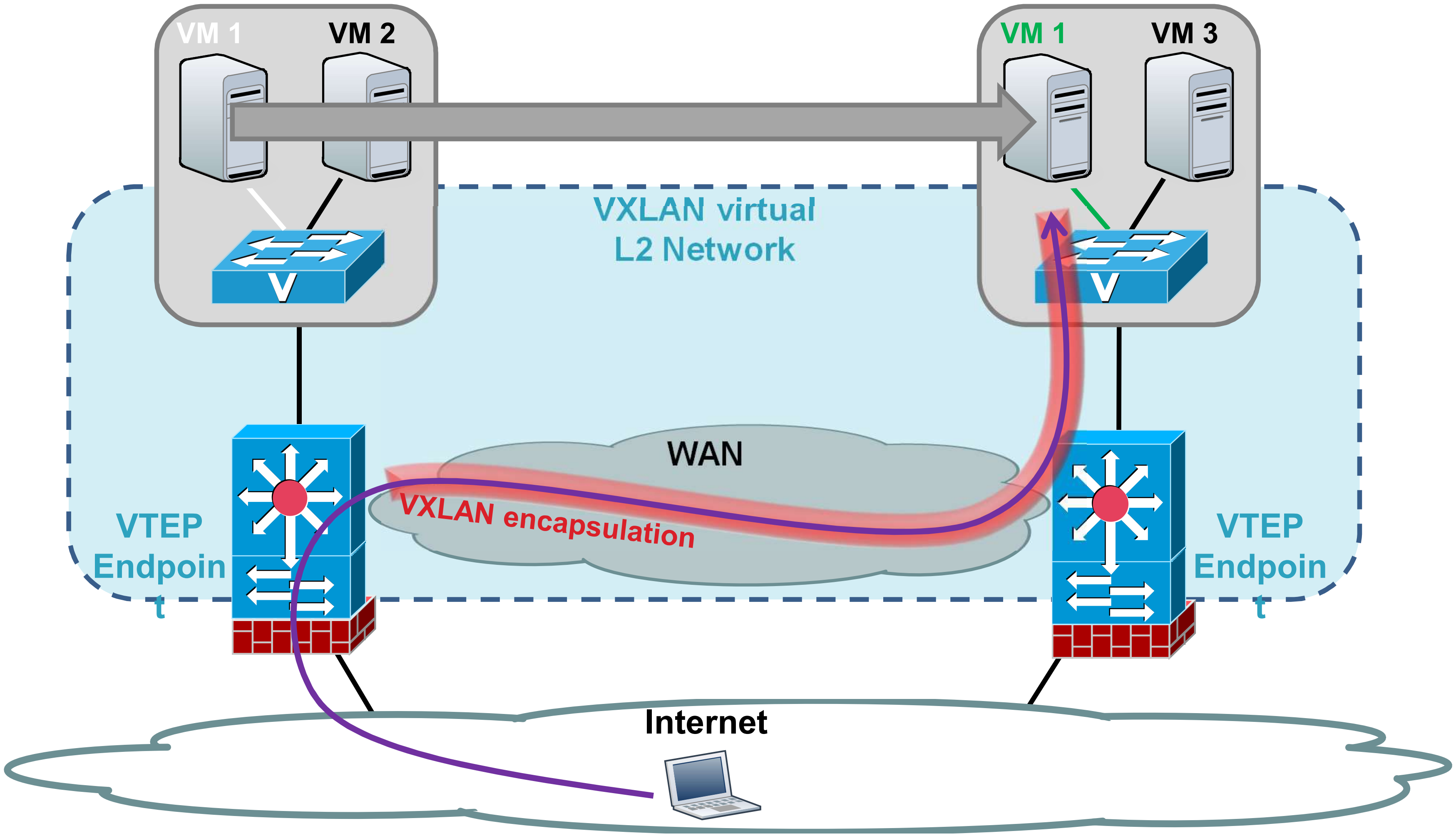



**Migration on different L3 network:**

- Need:
  - New VM's IP config
  - DNS config
- Problems when using:
  - Load balancing
  - Redundancy
- …
- Need IP reconfiguration

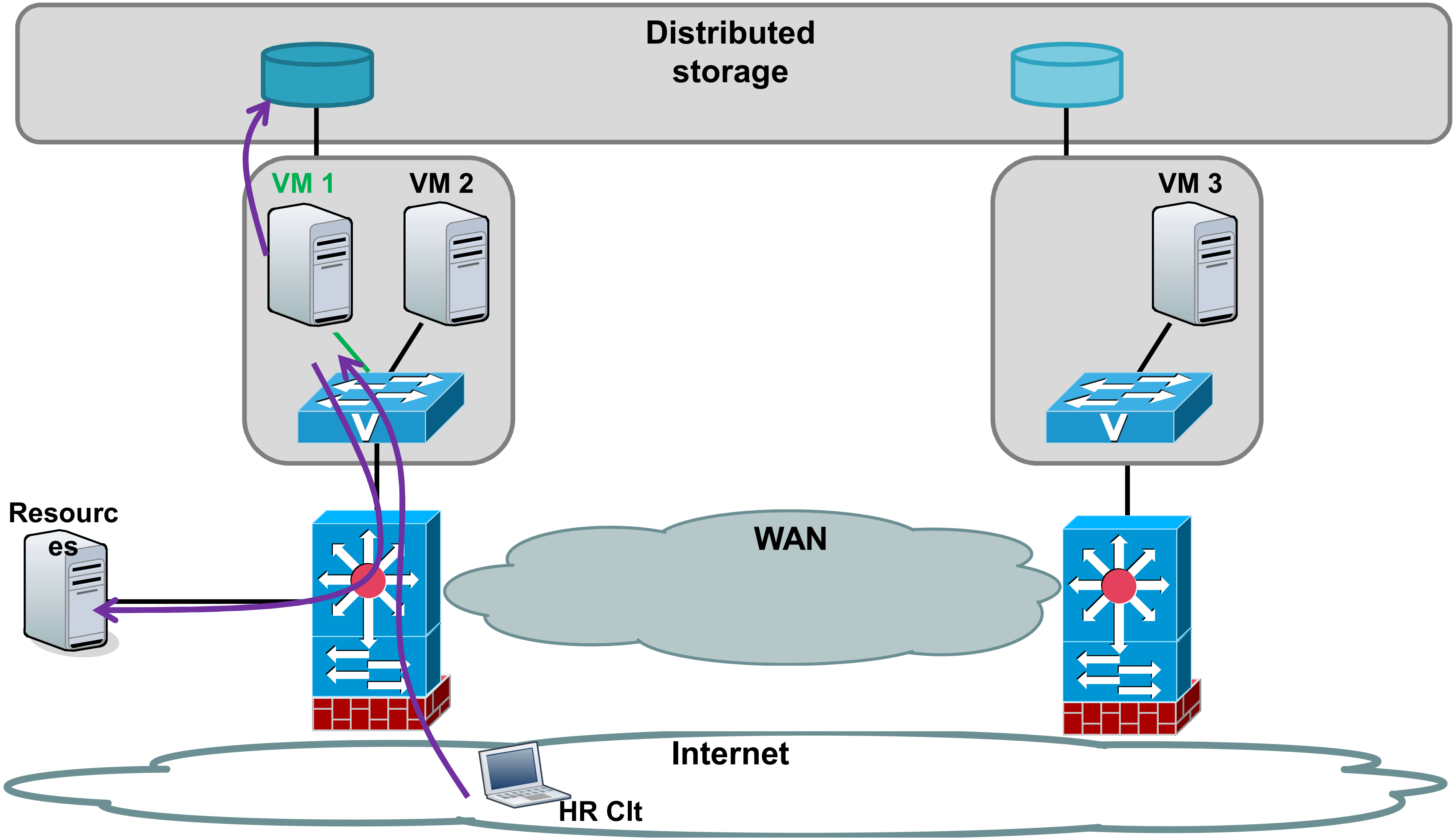
# VM Migration – VXLAN

## VM Migr. with VXLAN (Same principle for NVGRE or STT)



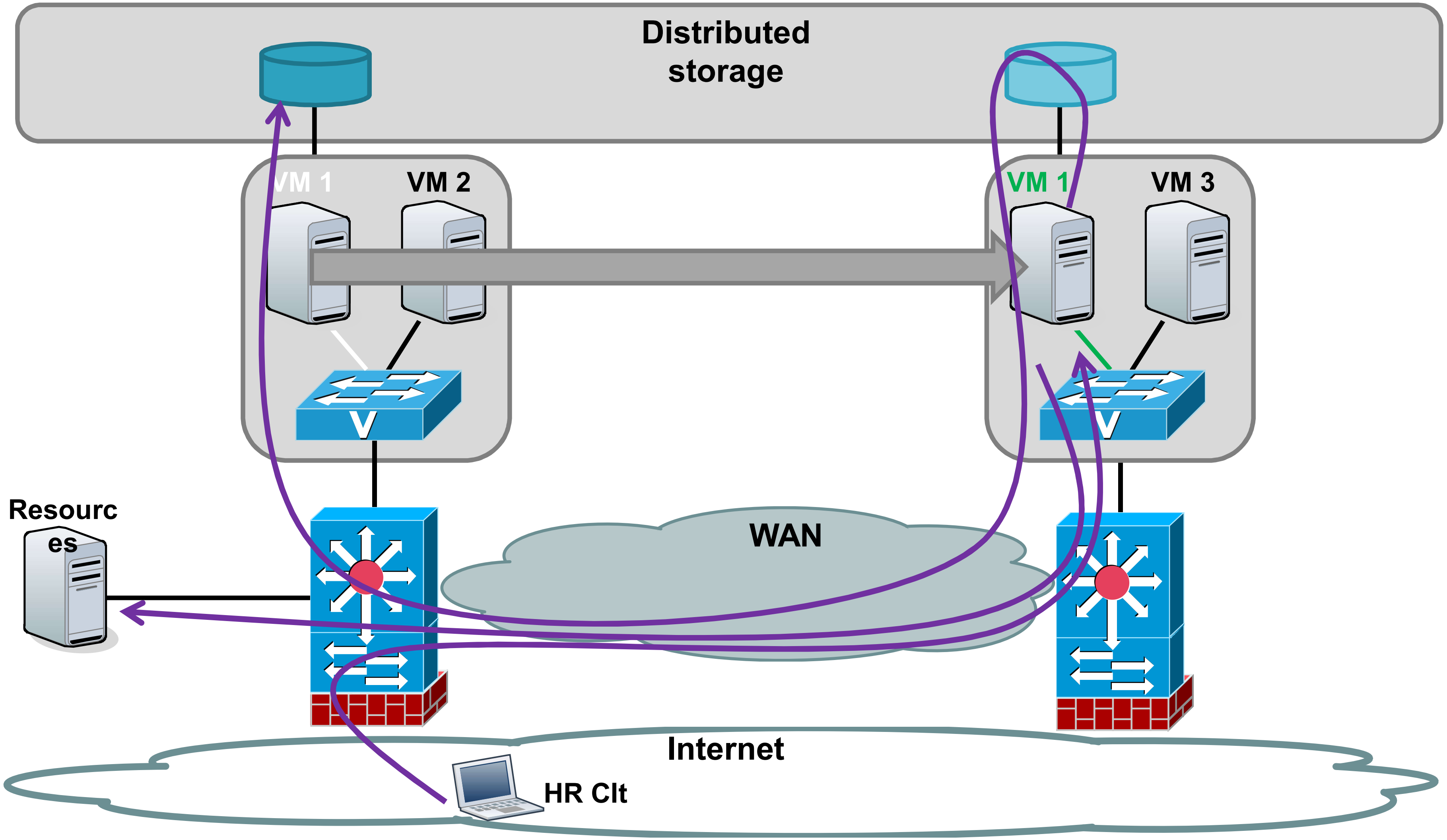
# VM Migration – Traffic Trombone – 1

## Before VM migration



# VM Migration – Traffic Trombone – 2

## After VM migration



# VM Migration

**«Network's flexibility needs to be increased and this can be done with Software Define Network (SDN)»**

# VM Migration – Glossary

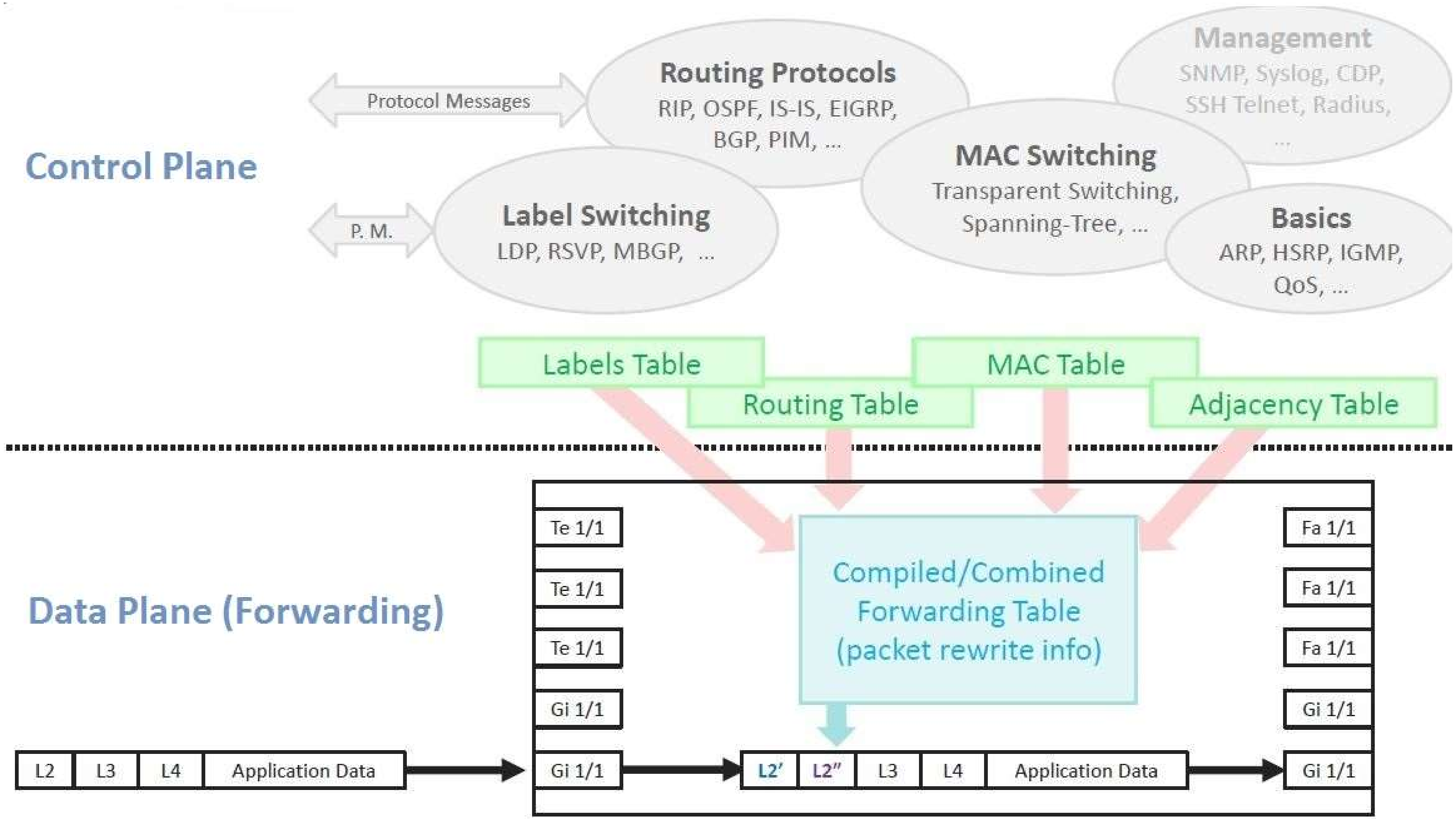
## References

- https://www.vmware.com/products/vsphere/vmotion.html
- https://www.vinchin.com/vm-migration/vmware-vmotion.html
- http://searchnetworking.techtarget.com/feature/Long-distance-vMotion-traffic-trombone-so-why-go-there
- http://searchnetworking.techtarget.com/tip/Live-migration-networking-Enter-long-distance-data-center-bridging
- http://searchnetworking.techtarget.com/tip/vSphere-vSwitch-primer-Design-considerations
- https://forum.proxmox.com/threads/migrating-vm-from-vmware-to-proxmox-and-retaining-network-config.143715/
- https://proxmox.com/en/

# SDN - Definition

- «Software Defined Networking is an emerging network architecture/concept where network control (control plane) is decoupled from forwarding (data plane) and is directly programmable.»
- In network devices, we can distinguish :
  - **Control planes :** represent decision and directly influence network behaviour
  - **Data planes :** (forwarding) represent forwarding decision taken on data
- SDN is a centralized control plane architecture which offers a global view of the network to applications.
- … but before presenting SDN more in details, what is a traditional network device?

# SDN – Trad. Network Devices



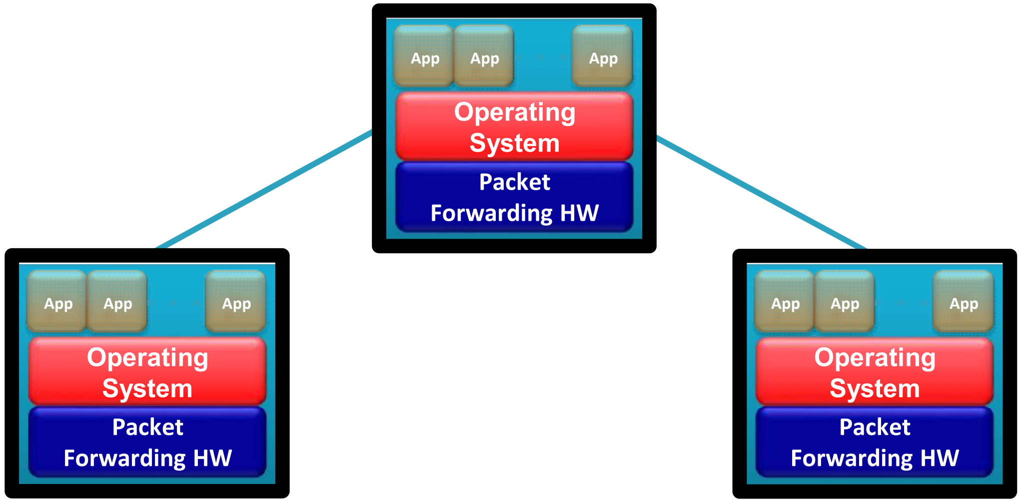

**Traditional network device representation**

# SDN – Limits of Trad. Networks

- Traditional networks offer an abstraction model in layers (OSI Model) and a protocols functionality division implying:
- **Specificity**: Each protocol solves a specific problem without profits of any fundamental abstraction (each one with its specific exchange/table)
- **Communication**: Protocol implies devices communication to grab information on global network topology
- **Configuration**: Each device needs to be separately configured implying management complexity
- **Flexibility**: Change in network architecture imply many devices configuration
- *Example: OSPF is a network routing protocol which exchanges information between devices to construct a global network topology and produces a forwarding table (Control Plane). This table is created by each device and used to take the best routing/forwarding decision (Data Plane). Each device need to be configured to use OSPF.*

# SDN – Trad. Network Architecture

- In traditional networks :
  - Control is distributed
  - Data/Control Planes are combined
  - Devices are closed source
  - Interactions between manufacturers are difficult
  - Each switch is configured independently
  - Distributed protocols between switch to share network information
  - A protocol per specific problem

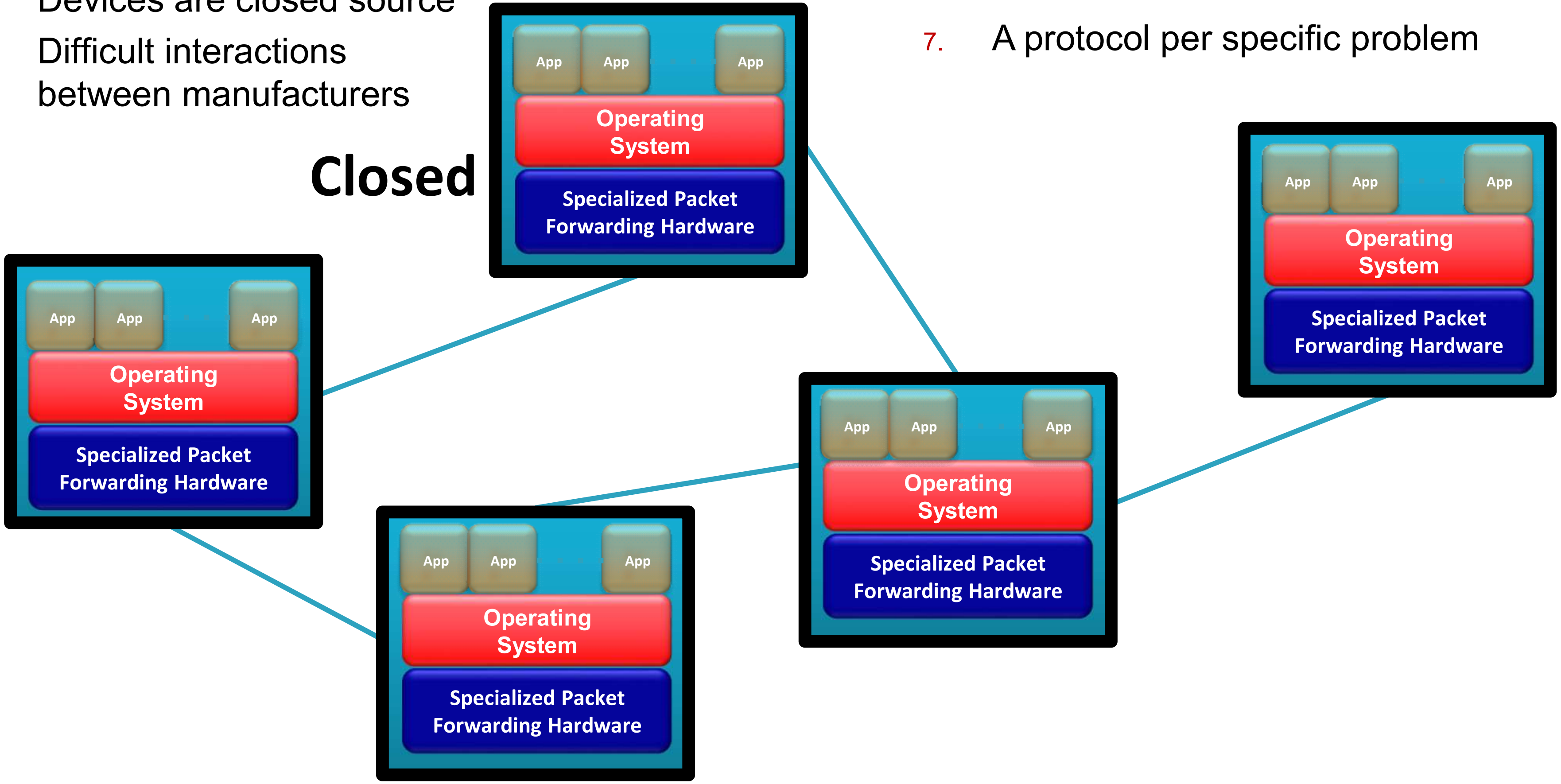



**OpenFlow/SDN tutorial, Srini Seetharaman, Deutsche Telekom**

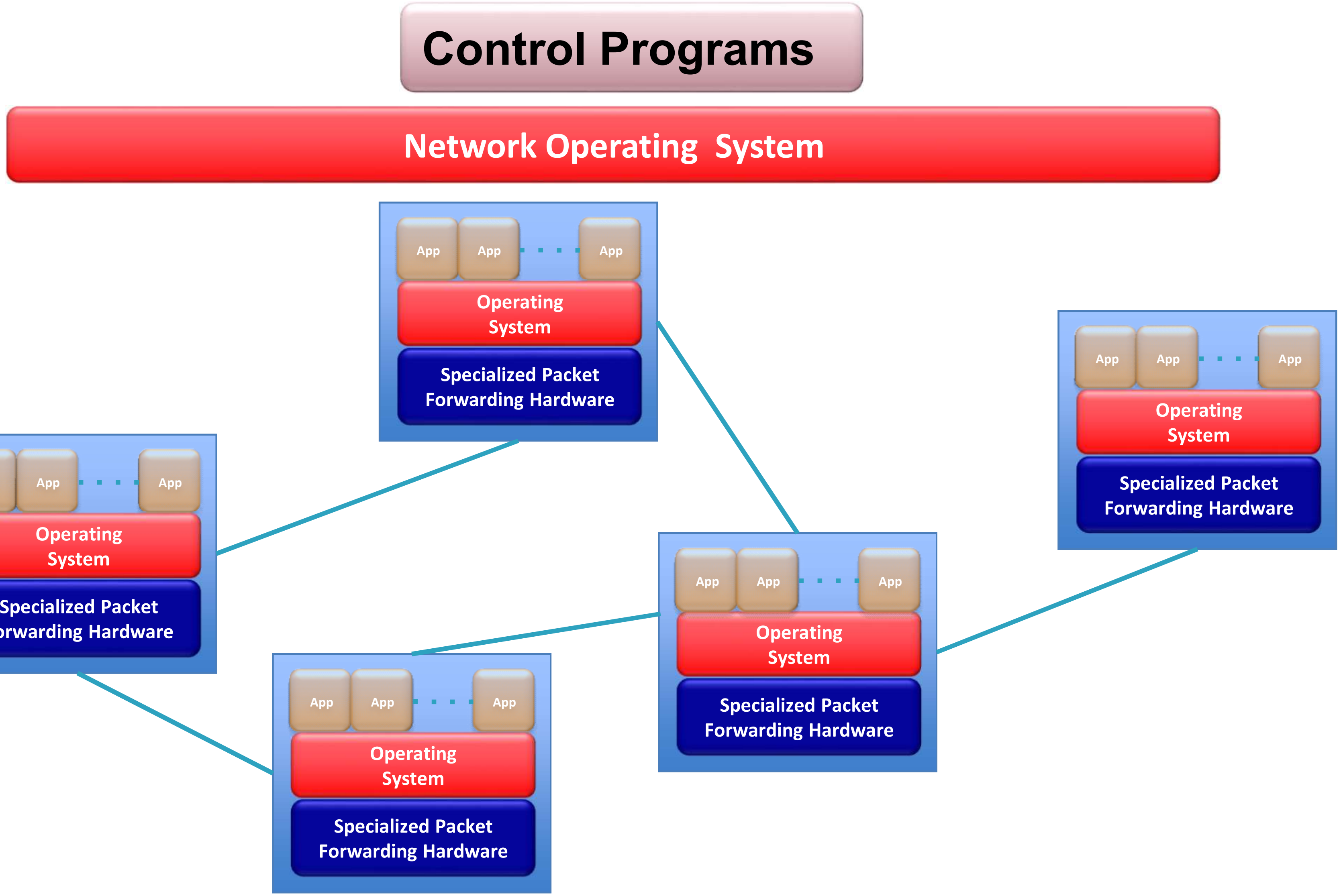
# SDN – Trad. Network Architecture

## *In Traditional Networks:*

1. Control is distributed
2. Data/Control Planes are combined
3. Devices are closed source
4. Difficult interactions between manufacturers
5. Each switch configured independently
6. Distributed protocols between switch to share network information
7. A protocol per specific problem



# Idea: An OS for Networks



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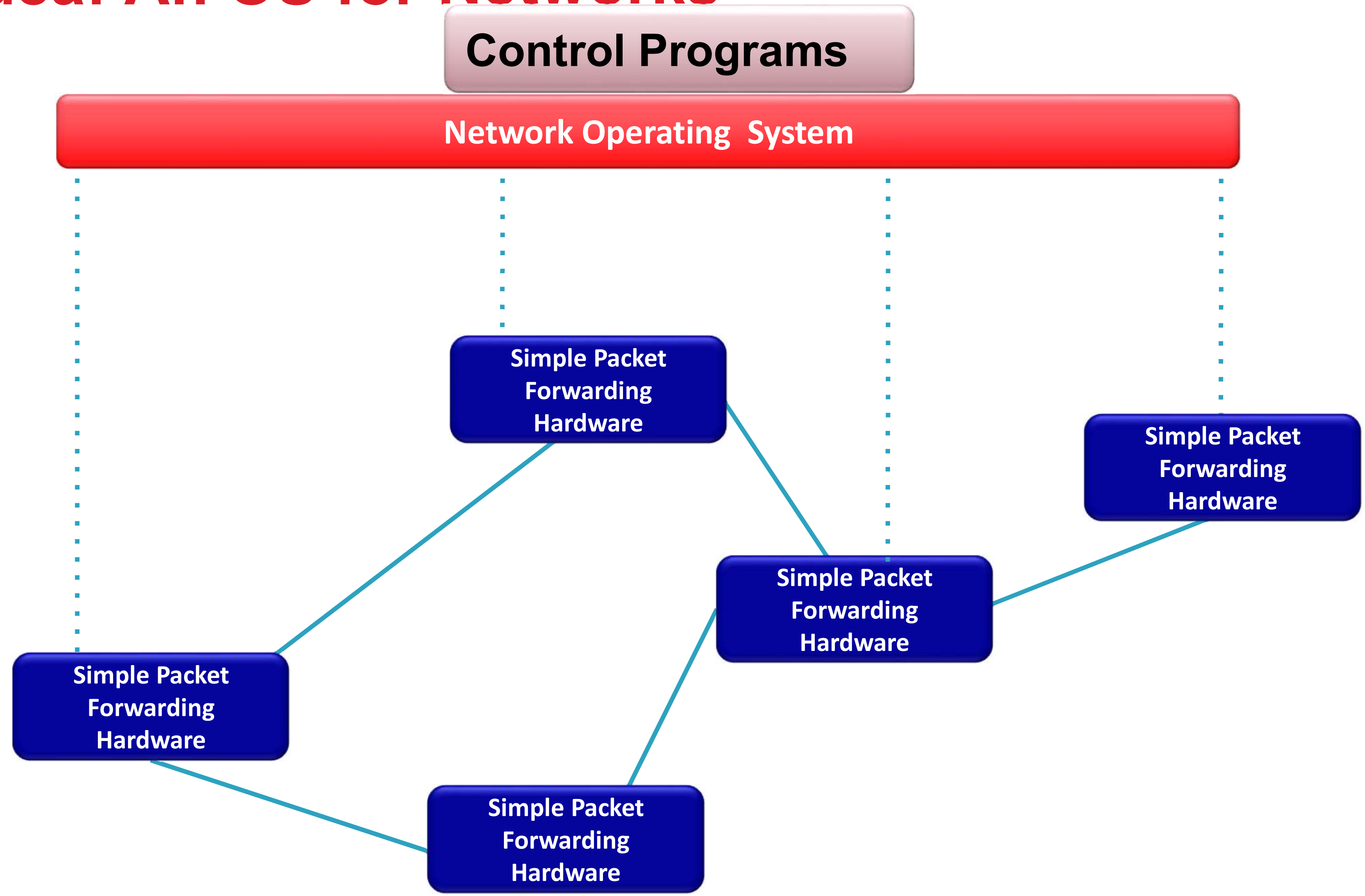




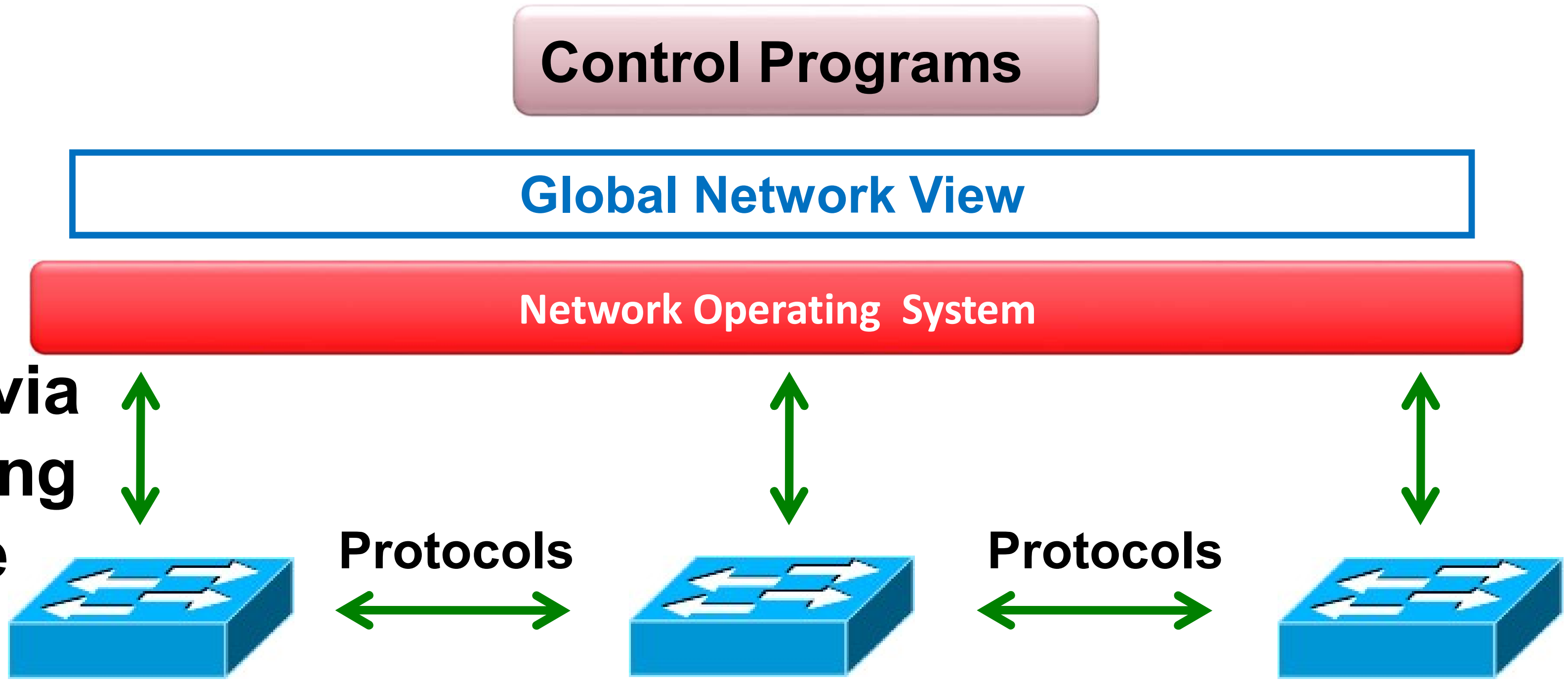

OpenFlow/SDN tutorial, **Srini Seetharaman, Deutsche Telekom, Silicon Valley Innovation Center**

# Idea: An OS for Networks

- "NOX: Towards an Operating System for Networks"

**Software-Defined Networking (SDN)**

Control via forwarding interface

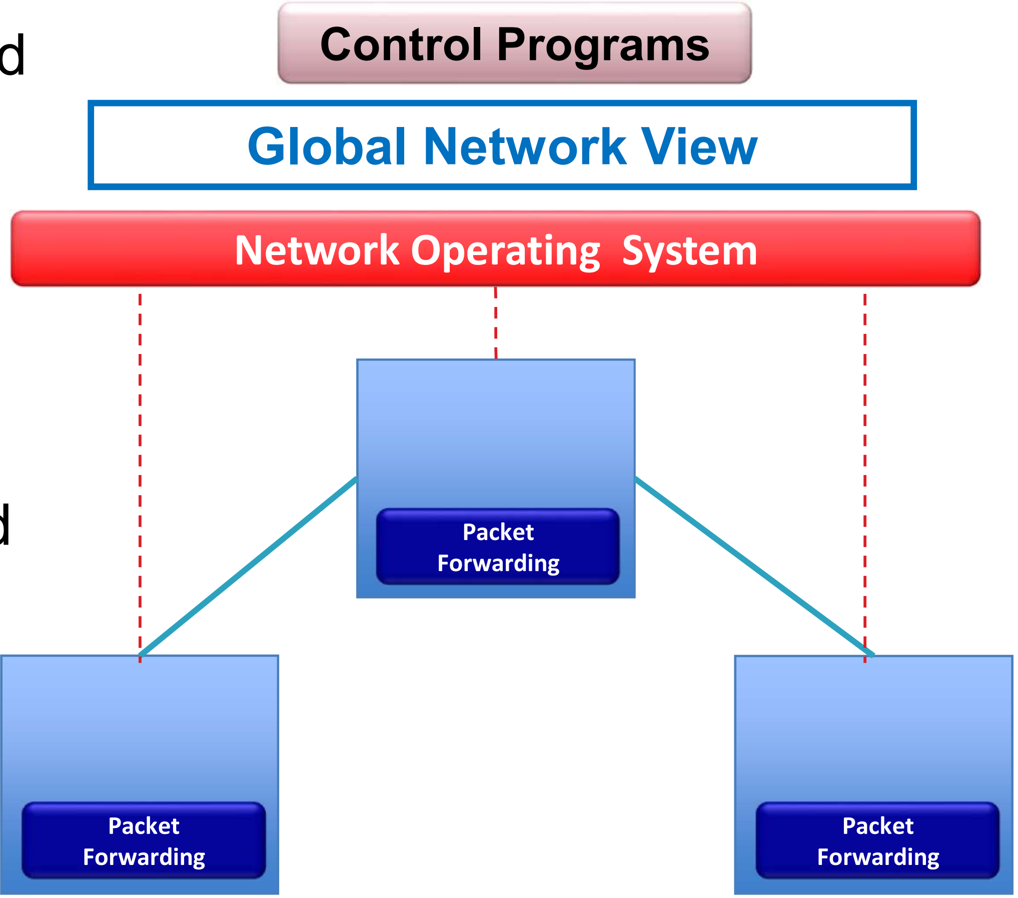





The Future of Networking, and the Past of Protocols, Scott Shenker, *with Martin Casado, Teemu Koponen, Nick McKeown*

# SDN – SDN Network Architecture

- In SDN network :
  - Control/Data planes are separated
  - Control (brain) is centralized in a Network OS which:
    - Gathers information from network elements
    - Offer a global network view
    - Disseminate control command
  - Control programs exploit the global network view to offer functionalities like routing, access control, etc.
  - Devices do only forwarding actions



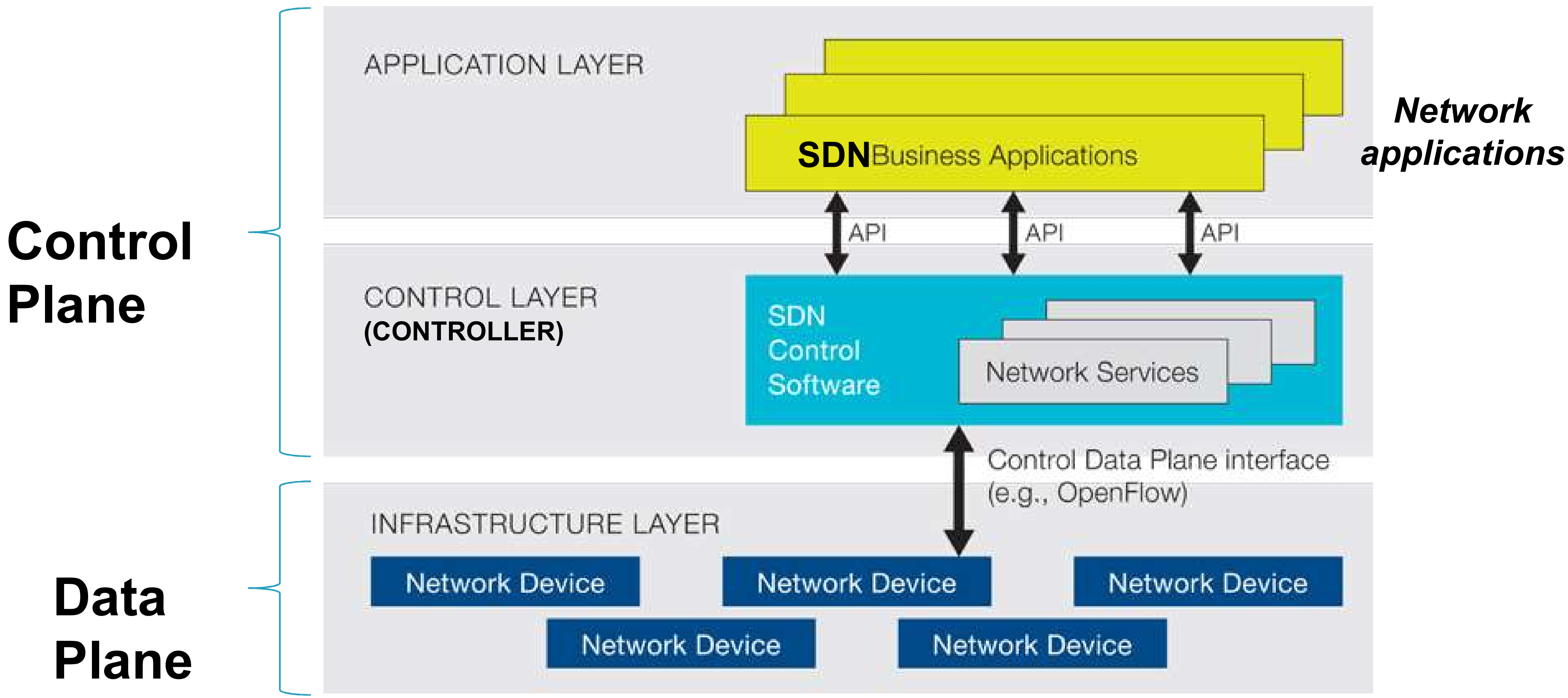

OpenFlow/SDN tutorial, Srini Seetharaman, Deutsche Telekom

# SDN – Benefits

- Configuration is simplified because it is centralized
- Architecture is dynamic/adaptable
- Everyone can write application, not only manufacturers
- Devices can be more simple, fast and inexpensive
- Programing interface improve :
  - Verification
  - Extensibility
  - Maintainability

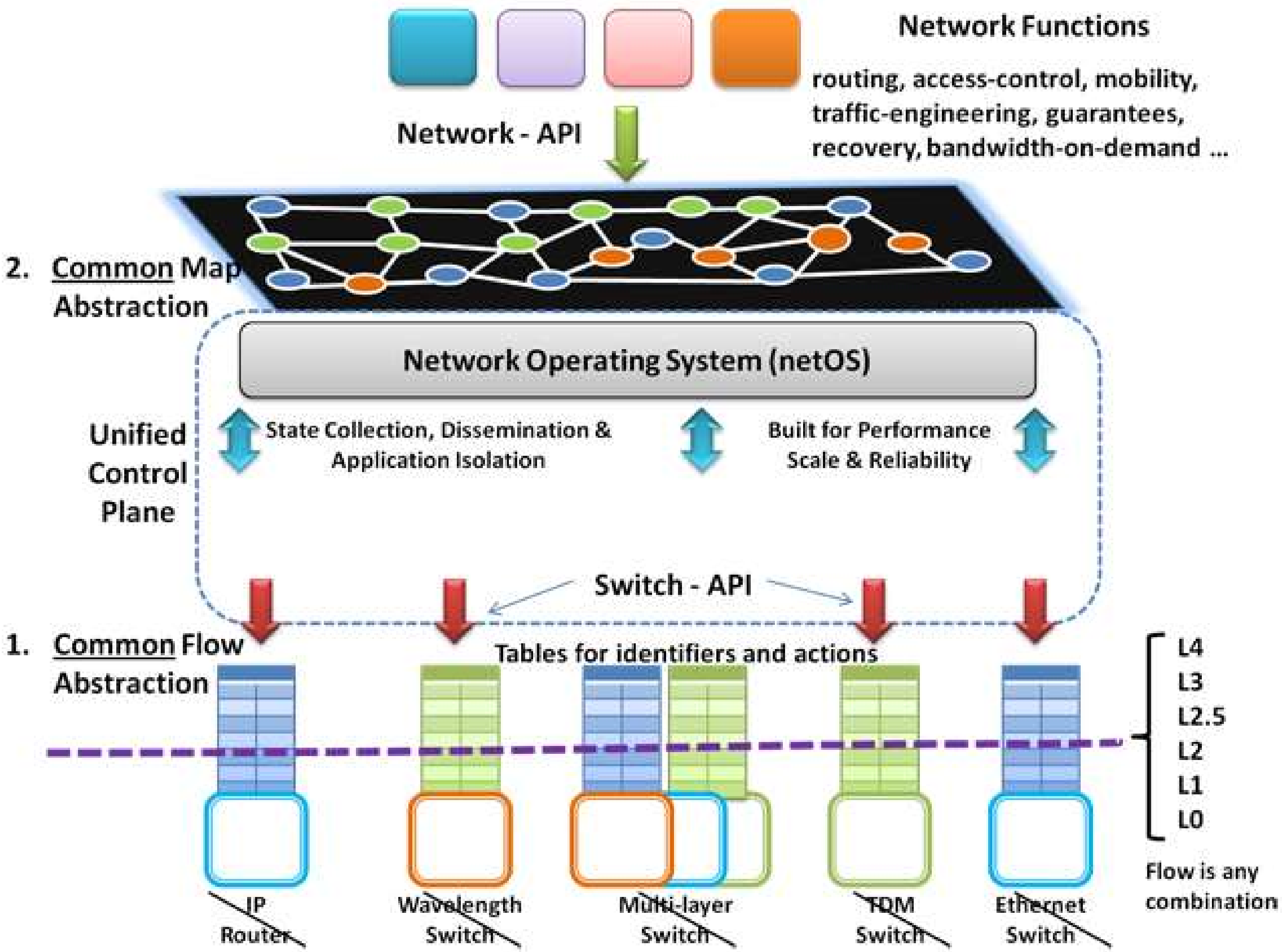
# SDN – Rise of SDN

# SDN – Detailed Architecture – 1




**Logical view of SDN Architecture**

# SDN – Architecture – 1



# SDN – Architecture – 2

- SDN Elements include:
  - **Applications**: Network programs which use controller API to define network/devices behaviour
  - **Northbound API**: Interface between applications and controller. Allows application to program the network for their needs. (ex: REST, Java, NETCONF, XMPP…)
  - **Controller**: Software element that abstract network for upper layer. Controller is the core of SDN networks. Any communications between applications and devices have to go through the controller
  - **Southbound API**: Interface between controller and devices. Used to collect information from devices and disseminate control commands (ex: Openflow, ForCES, I2RS/BGP-LS, OKE-PK, XMPP ..)
  - **Devices**: Move packets between interfaces

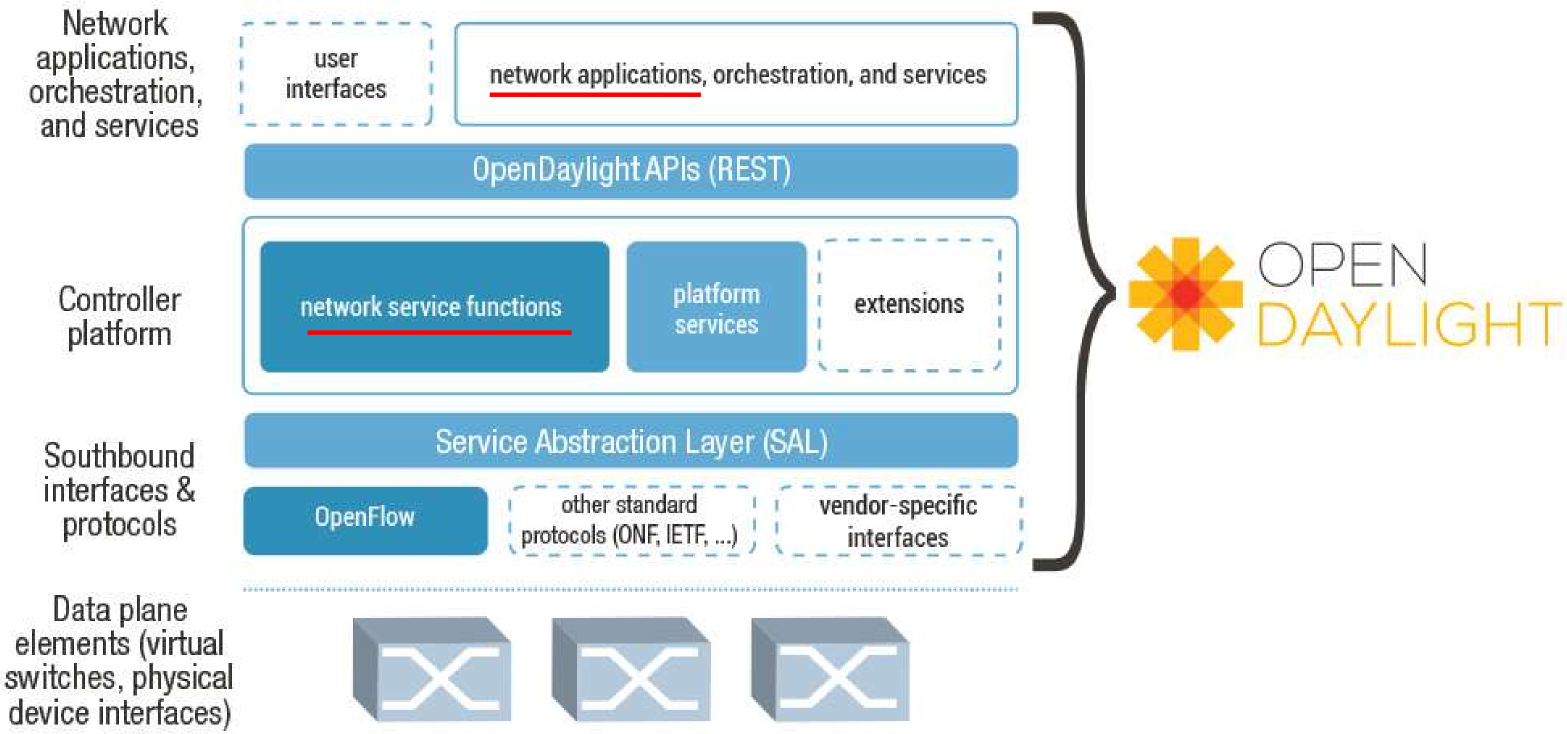
# SDN – Controllers – 1

- A lot of new solutions based on SDN concept exist
- Existing controller:
  - Floodlight (OpenSource)
  - Big Network Controller (OpenSource)
  - Cisco ONE SDN Controller (Proprietary)
  - HP Virtual Application Networks SDN Controller (Proprietary)
  - Juniper JunosV Contrail Controller (Proprietary)

- … Each controller uses different Northbound API implying incompatible applications

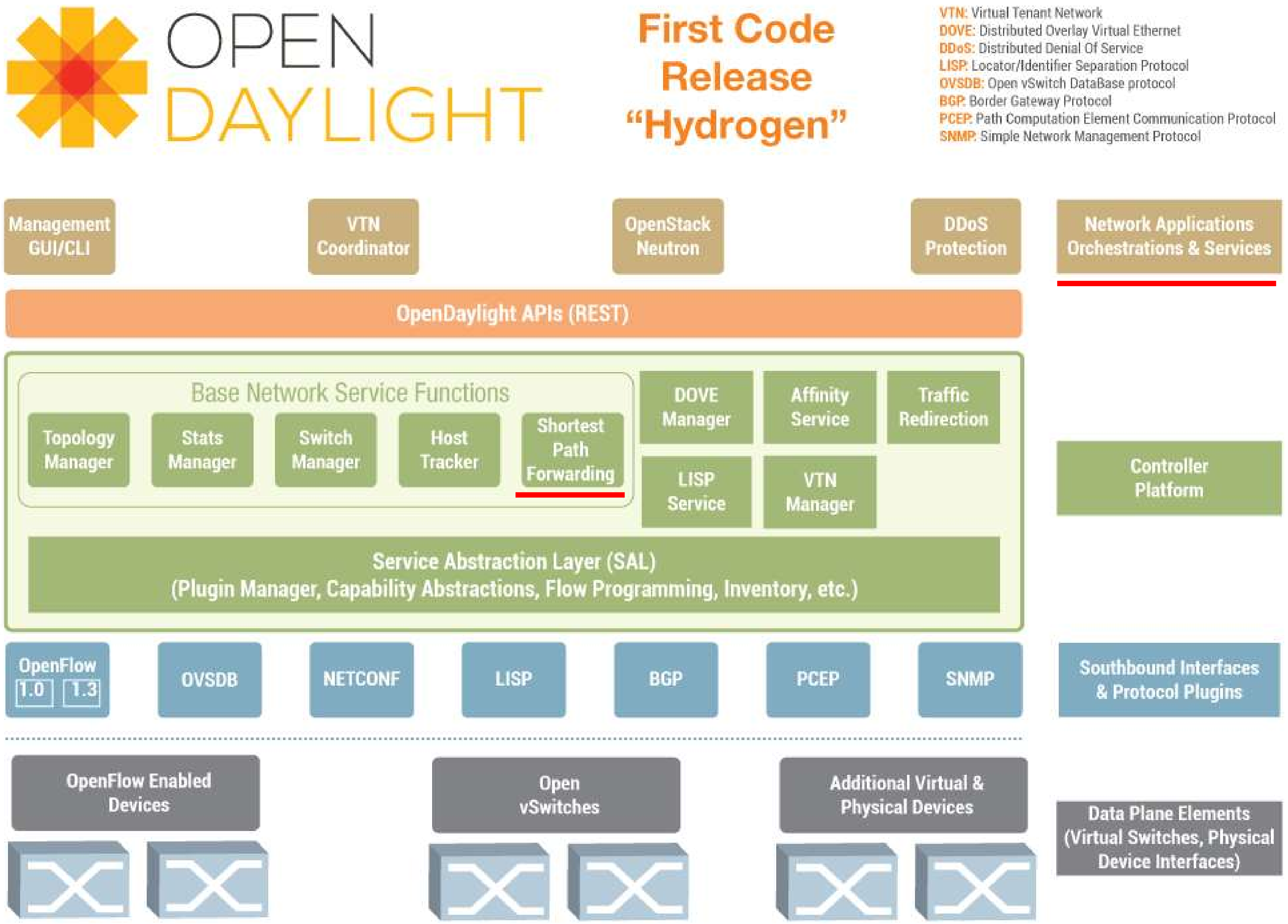
# SDN – Controller – 2

- OpenDaylight is an open source project
- Aims to create a common open source SDN Controller and define a reference framework for applications
- Based on Cisco ONE SDN Controller
- First release in Q4 2013, version 4 available now

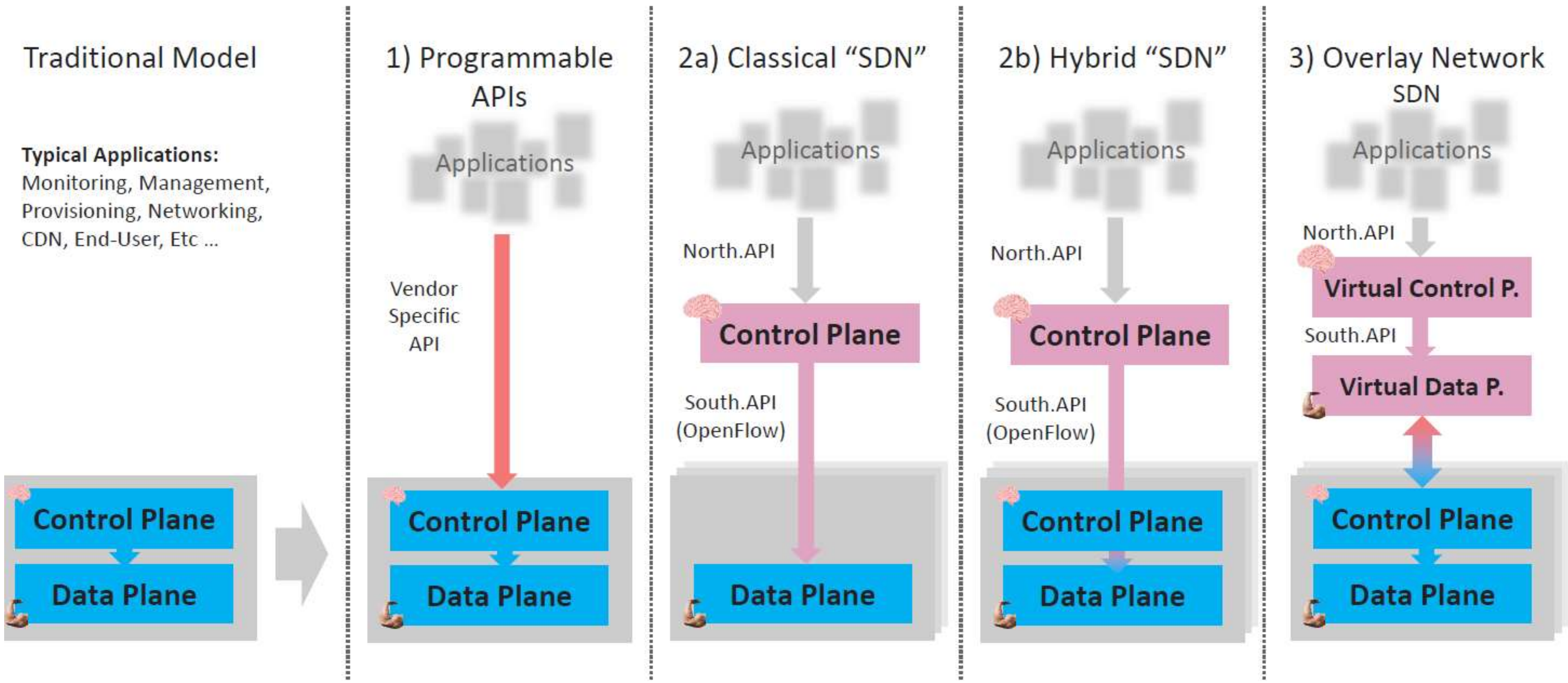


# SDN –Controller – 3 (details)

- Application Layer
- Control Layer
- Data Plane

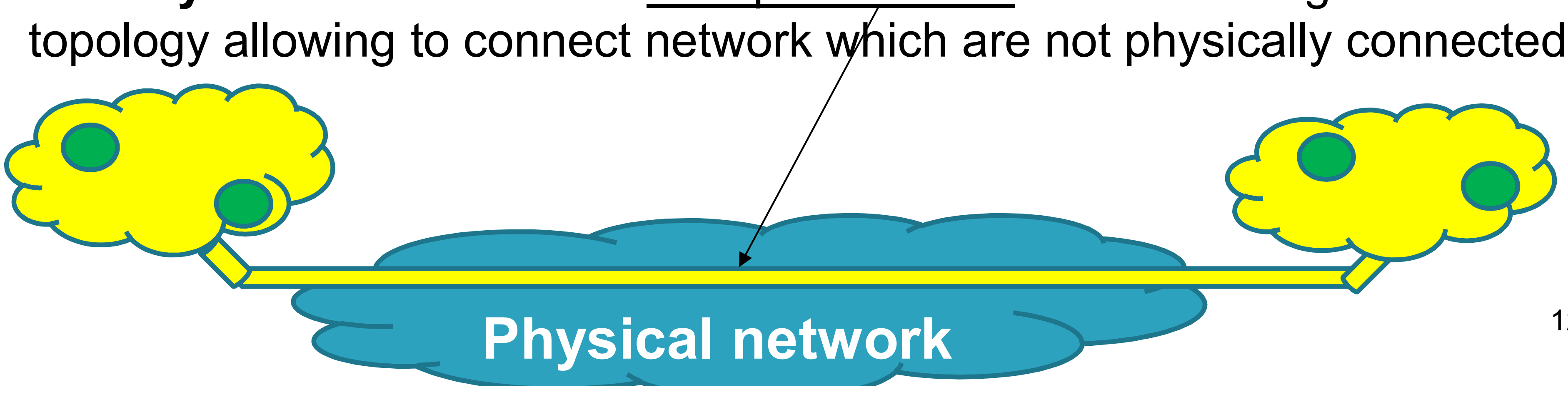


# SDN – Overlay Network – 1



# SDN – Overlay Network – 2

- Definition:
  - **Traditional Network**: traditional device which merge data plane and control plane
  - **Programmable API**: Proprietary applications and interface to simplify network management of traditional devices
  - **Classical SDN**: SDN only network with centralized control plane
  - **Hybrid SDN**: Hybrid switch which could use internal control plane (manufacturer) or centralized
  - **Overlay Network SDN**: Use data plane tunnel to create a logical topology allowing to connect network which are not physically connected



# SDN – Some examples of Current Solutions

**Juniper MX-series**

**NEC IP8800**

**WiMax (NEC)**

**HP Procurve 5400**

**Netgear 7324**

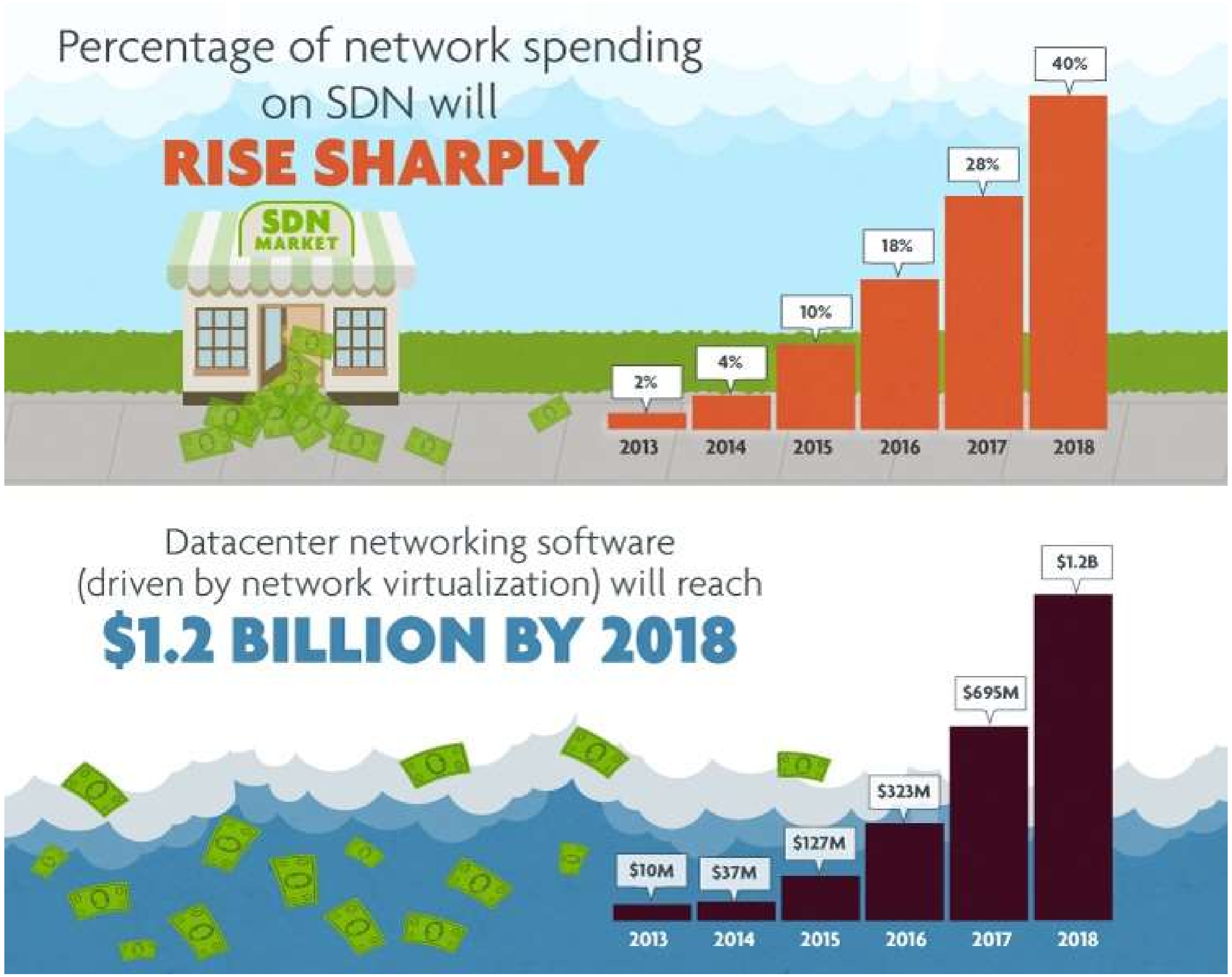
**PC Engines**

**Pronto 3240/3290**

**Ciena Coredirector**

**And many others...**

# SDN – SDN market forecast – 2



# SDN – SDN market forecast – 3

# SDN – SDN market forecast – 4

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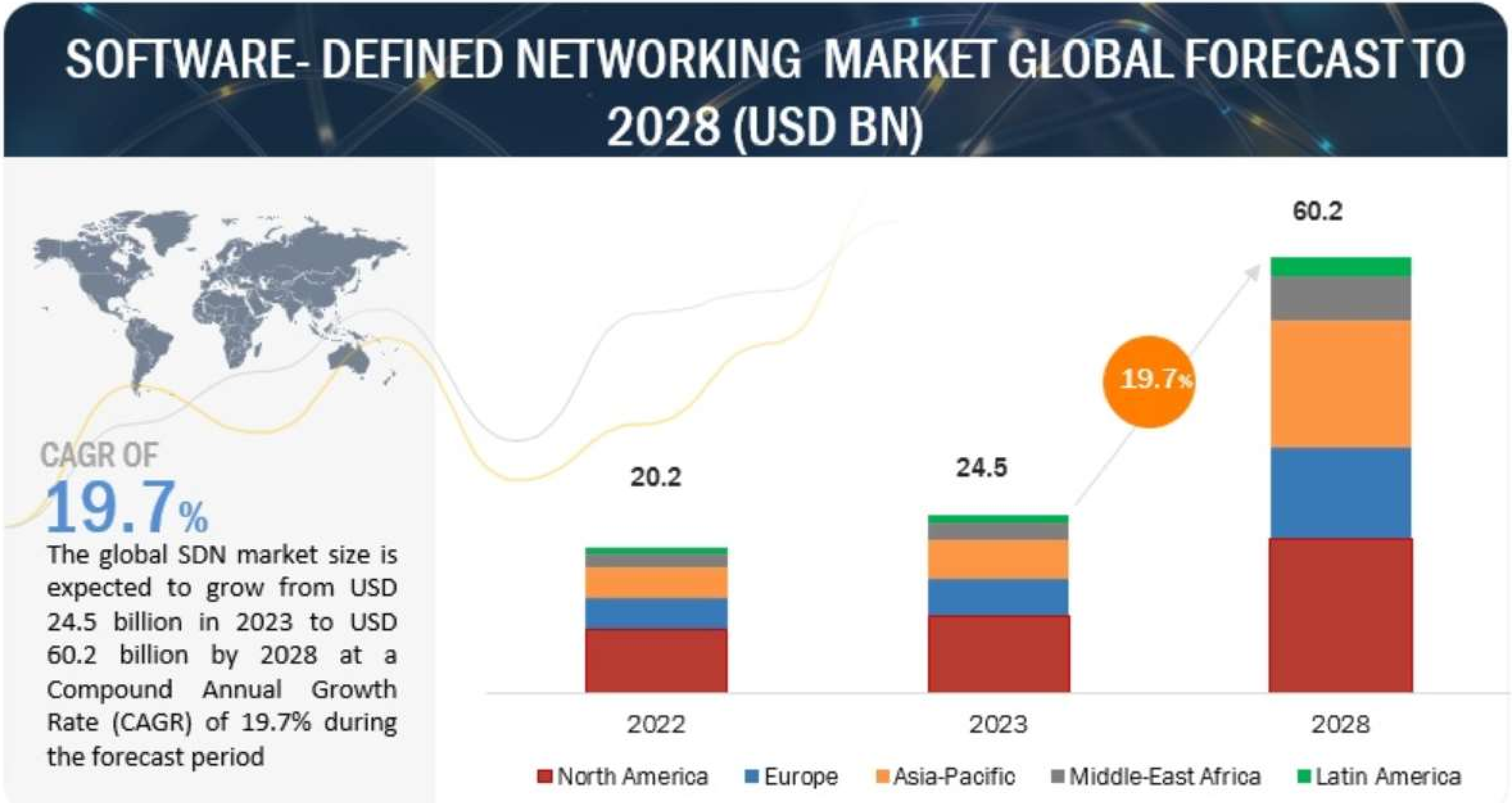
# SDN – SDN market forecast – 1

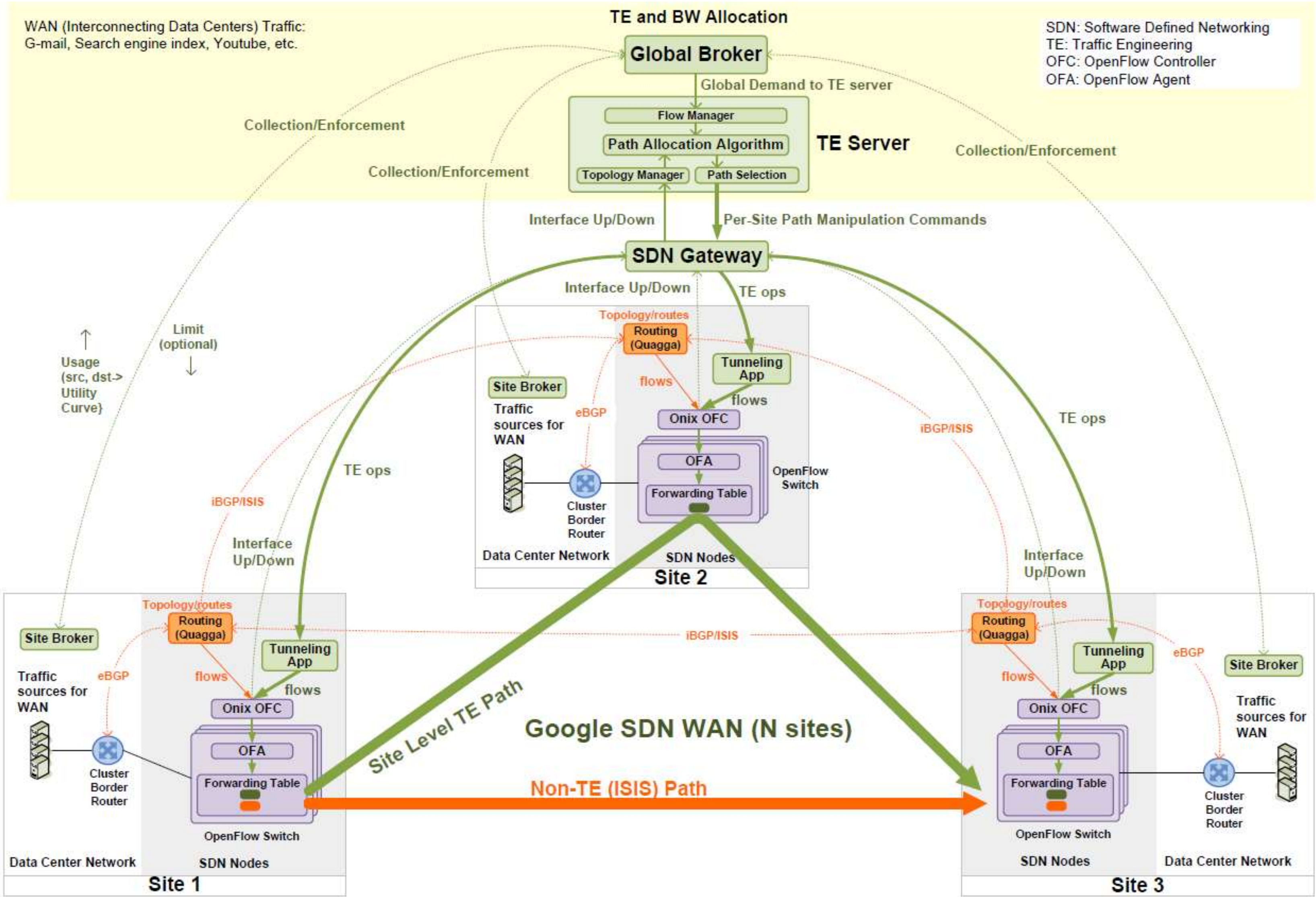



https://www.marketsandmarkets.com/Market-Reports/software-defined-networking-sdn-market-655.html?gad_source=1&gclid=Cj0KCQjw0MexBhD3ARIsAEI3WHL1-tmJ_OdHD-XDDPSVihCQ51Exmq-KT8NvIxraJlfXbKcuFWWSHOAaAr4wEALw_wcB

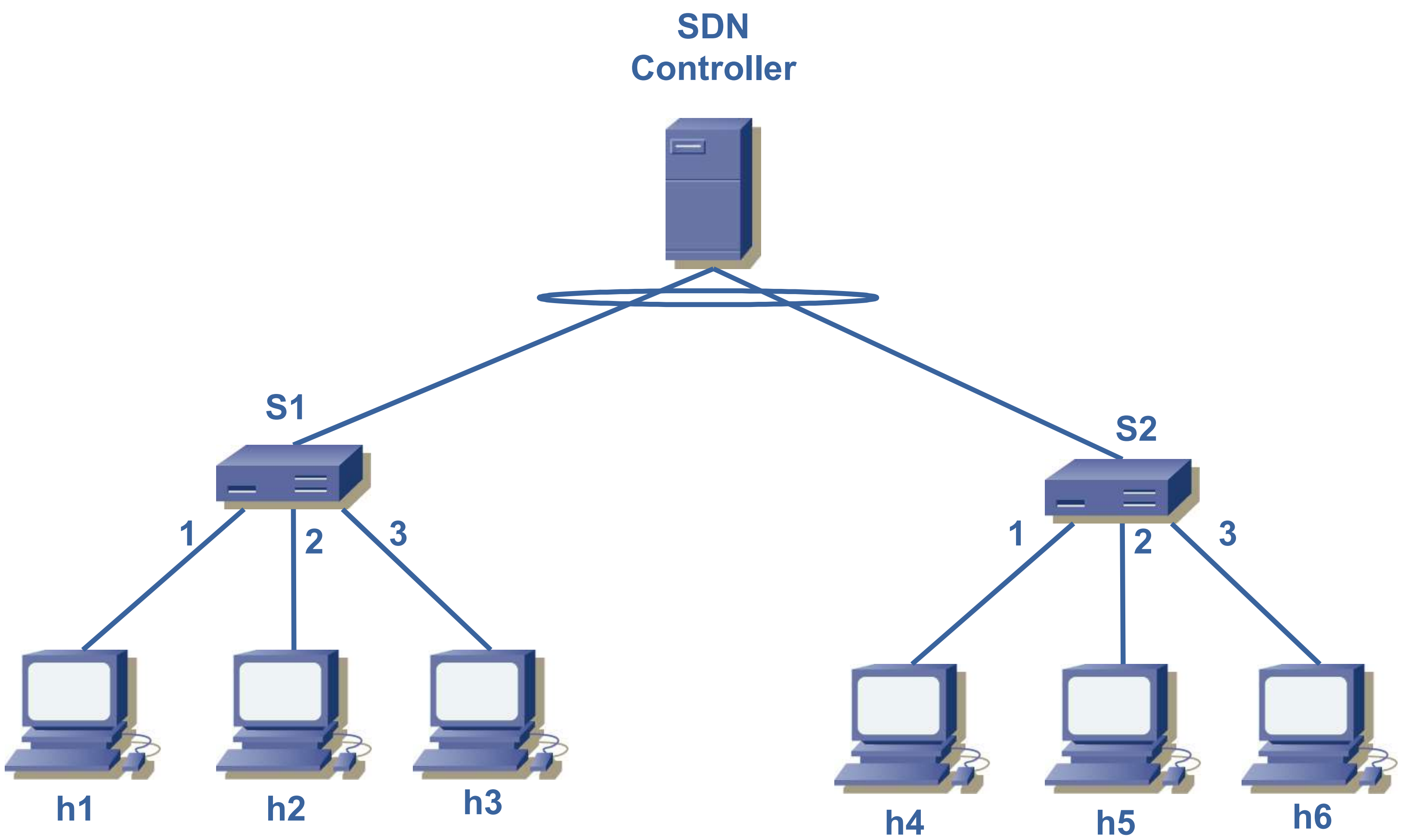
# SDN – Google SDN WAN – 1

- Infrastructure SDN sample: Google SDN WAN
- Why SDN for Google?
  - Separate hardware from software
    - Choose hardware based on necessary features
    - Choose software based on protocol requirements
  - Logically centralized network control
    - More deterministic
    - More efficient
    - Fault tolerant
  - Automation: Separate monitoring, management, and operation from individual boxes

# SDN – Google SDN WAN – 2



# Exercice: understanding SDN integration



# SDN – Glossary

## References

- https://www.opennetworking.org/sdn-resources
- http://www.opennetsummit.org
- https://www.netmanias.com/en/
- https://sdn.systemsapproach.org/netvirt.html
- https://cloud.google.com/blog/products/networking/google-global-network-technology-deep-dive